Entrepreneurship and Intellectual Property Rights

HS351TA | V Semester SEE Question Papers

- Feb/Mar 2026 (HS351TA)
- Feb/Mar 2025 (HS351TA)
- Apr/May 2024 (21HS51A)
- Apr 2024 (18HSI51)
- Jun 2023 Makeup (18HSI51)
- Mar/Apr 2023 (18HSI51)
- Apr 2023 (16HSI51)
- Jan/Feb 2021 (18HSI51)
- Jan/Feb 2021 (16HSI51)
- Nov/Dec 2019 (16HSI51)
- Fast Track Jul 2019 (16HSI51)

Question Paper ID: 315151

Course Code: HS351TA

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

Common to AS / CH / CV / EC / EE / EI / ET / IM / ME

Entrepreneurship and Intellectual Property Rights

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Identify one analytical method used for opportunity identification.	02	1	1
1.2	Differentiate between an entrepreneur and an intrapreneur.	02	1	1
1.3	State the main advantage of cross-disciplinary collaboration for technological innovation.	02	2	2
1.4	What is meant by Entrepreneurial Finance and Resource Management?	02	3	3
1.5	Define the 4Ps in entrepreneurial marketing	02	1	1
1.6	Identify the purpose of Targeting in the STP process.	02	2	2
1.7	Mention any two objectives of the Patent System.	02	4	2
1.8	Mention any two methods used for IP Valuation.	02	4	2
1.9	What is Patent Drafting?	02	4	2
1.10	Mr. X has discussed an idea of video games with Mr. Y and Mr. Z. Later, Mr. Y has written a Video games based on discussion. Who will be author and owner of video games?	02	1	1

Part B

Question No	Question	M	CO	BT
2a	Discuss the role of entrepreneurship in promoting innovation and contributing to economic development. Support your answer with real-world examples.	08	1	3

2b	Differentiate between innovative, imitative, and Fabian entrepreneurs with suitable examples. Analyze how each type contributes differently to industrial and economic growth	08	1	4
3a	Define Market Opportunity and explain the key factors to identify potential market opportunities.	08	2	2
3b	Evaluate the advantages and limitations of growth strategies such as mergers, acquisitions, and strategic alliances.	08	2	2
	OR			
4a	Evaluate the importance of technical feasibility through prototype development and proof of concept.	08	2	2
4b	Design a business model canvas for a mobile app providing real-time traffic and parking solutions.	08	3	4
5a	Develop a marketing mix (4Ps) for a newly launched technology-based startup product.	08	1	3
5b	Construct an STP (Segmentation, Targeting, Positioning) model for a startup entering a competitive consumer market.	08	2	4
	OR			
6a	Formulate a Unique Value Proposition (UVP) and branding strategy for a product aimed at millennial customers.	08	2	3
6b	Analyze how CRM tools can enhance customer retention and long-term sales growth in entrepreneurial ventures.	08	1	4
7a	Discuss in detail the concept, scope, and salient features of patents. Explain with examples the kinds of inventions that are patentable under the Indian Patent Law. How does understanding these distinctions help in promoting innovation?	08	4	3
7b	Define “Patent Transfer”. Discuss the various methods by which patent rights can be transferred.	08	4	3
	OR			
8a	Define Eco-Label. How does it differ from a Trade Mark?	08	4	3
8b	Explain the different types of Trade Mark Transfers Under the Act.	08	4	3
9a	Explain the acts which do not constitute Infringement of Copyright.	10	3	4
9b	List the advantages of protecting a design under Industrial design right.	06	2	2
	OR			
10a	Discuss any five works that can be claimed under copyrights.	10	1	1
10b	Identify and explain the key tools and measures available in India to protect trade secrets effectively?	06	2	2

HS351TA

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Regular Examinations Feb/March-2025

COMMON TO AS / BT / CH / CV / IEM / ME

ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A

			M	BT	CO
1	1.1	List four types of entrepreneurs.	02	1	1
	1.2	Define: Entrepreneur.	02	2	1
	1.3	Identify the Activities involved in Market Demand Assessment.	02	4	2
	1.4	How can a business plan help in risk management?	02	4	2
	1.5	Explain the two types of promotional strategies.	02	2	3
	1.6	_____ is a fundamental framework in marketing that helps businesses identity and reach their most valuable customers.	02	1	3
	1.7	Give an example which brings about the fact that a mark cannot be registered as trademark if it consists exclusively of goods.	02	4	5
	1.8	List any two different forms of Trademark (Any Two).	02	1	5
	1.9	What are the criteria for a design to qualify for registration as an industrial design?	02	2	4
	1.10	Define performer's Rights?	02	3	5

PART-B

2	a	Justify the importance of entrepreneurship.	08	5	1
	b	What factors contributed to Flipkart's success in the Indian market.	05	4	1
	c	What challenges did Flipkart face during its growth phase?	03	3	1
3	a	A company's gross profit margin is increasing, but its net profit margin is decline. What could be the potential reasons?	08	4	2
	b	List the key Contents of a Business Plan (Any 8)	08	1	3
		OR			
4	a	Explain the key components of financial feasibility analysis?	10	2	2
	b	An entrepreneur wants to launch a niche product. What steps should they take to access market feasibility?	06	4	2
5	a	What is meant by Market Segmentation? Give its importance.	04	2	2
	b	A clothing brand wants to launch a new line targeting eco-conscious millennials. How should they approach market segmentation?	04	5	2
	c	Define *SWOC* Analysis. Explain its significance in strategic Management.	08	4	2

		OR			
6	a	What are the Types of Insurance as Risk Management tools?	08	2	2
	b	What are the objectives of financial management? Investigate financial management tools & Techniques**.**	08	5	2
7	a	Explain the tools to protect a trade secret?	08	1	4
	b	Explain the Inventions which are not patentable under the Indian Patent Act.	08	5	4
		OR			
8	a	Discuss the Trade mark infringement among the corporate classes in India with Two Case examples	08	2	5
	b	A company owns the trademark “Perfect Clean” for cleaning products. Over time, customers start using “Perfect Clean” as a generic term for all cleaning products. How can the company prevent the loss of its trademark rights?	08	5	5
9	a	List & explain steps involved for obtaining Design Protection under the *ID* Act.	08	1	4
	b	Copy Right protects expression but not the ideas. Discuss.	08	5	4
		OR			
10	a	A digital artist uses an AI tool to generate artwork and claims copyright over the output. Can they enforce copyright on this work?	06	5	5
	b	Explain the acts which do not constitute Infringement of copyright under the act.	10	1	4

21HS51A

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Regular Examinations April/May-2024

COMMON TO AIML/CSE/ISE/ECE/EEE/EI/ET

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

		PART-A	M	BT	CO
1	1.1	When the patentee agrees to give another person certain defined right in the patent with immediate effect by a document and not by an agreement the assignment is termed as _____.	01	2	1
	1.2	An assignment of an existing patent through an _____ which has been duly registered is called legal assignment.	01	1	2
	1.3	Whether the assignee of a true inventor can ask for Patent?	01	4	1
	1.4	_____ of the Act makes it clear that the proprietor of a certification trademark should not himself carry on a trade in the goods of the kind certified or a trade of the provision of services of the kind certified.	01	1	1
	1.5	The term Entrepreneur is derived from the term “entreprendre” which means _____.	01	1	4
	1.6	Symbol of Maharaja of Air India is _____.	01	4	1
	1.7	What is the term for the act of registering someone else’s domain name with the intent to profit from its sale to the rightful owner?	01	1	2
	1.8	What does DRM stand for in the context of protecting intellectual property rights?	01	1	1
	1.9	How does watermarking contribute to protecting intellectual property rights in digital media?	01	4	2
	1.10	In India, the literary work is protected until _____.	01	1	1
	1.11	What symbol is commonly used to indicate copyright protection?	01	2	1
	1.12	What is the term used for gaining unauthorized access to a computer system or network?	01	2	2
	1.13	A student photographer takes a picture using a school-owned camera as part of a class assignment. The student never signed any agreement to give away his copyrights and the student handbook doesn’t say anything about who gets it. Who owns the copyright?	01	4	2
	1.14	_____ is an application of information and communication technology (ICT) for delivering Government Service.	01	1	1
	1.15	Susie’s department is implementing many projects. She finds herself starting and stopping work on one task to go and work on another task, and then return to the work on the original task. Susie is experiencing:	01	2	3
	1.16	_____ Monitors all internet and other network activity, looking for suspicious data and preventing unauthorized access.	01	2	2

	1.17	Which international agreement provides a framework for the protection of intellectual property rights, including in the digital environment?	01	4	2
	1.18	What is a subscription business model? Give an example.	01	1	4
	1.19	Give the classification of projects.	02	1	4

PART-B

2	a	What details should a complete specification contain?	08	1	1
	b	What is the Rice patent? Explain the Indian Government's response to the Rice patent.	08	2	2
3	a	Explain the Tools to protect a trade secret.	08	2	2
	b	Illustrate the Trade mark infringement/'passing off' case in the context of cookie manufacturer Malaysian.	08	3	2
		OR			
4	a	Illustrate and explain the different categories of NON-REGISTRABLE TRADEMARKS.	08	3	2
	b	Illustrate any two cases of Trade Mark Violation in India.	08	3	2
5	a	Describe the procedure for obtaining Design Protection.	10	2	2
	b	Illustrate any two case examples of your choice for Design Protection.	06	3	1
		OR			
6	a	Compare between Electronic Signature and Digital Signature (Any 8 differences).	08	3	2
	b	Illustrate the Infringement of Copy Right with suitable Case examples.	08	3	1
7	a	Analyze the different Entrepreneurial Development Models.	08	3	4
	b	Illustrate with appropriate examples myths of entrepreneurship.	08	3	3
		OR			
8	a	Explain the challenges encountered by the women in their entrepreneurial journey.	08	2	4
	b	Determine the measures required for Capacity Building for Entrepreneurship in India.	08	3	3
9	a	Analyze the reasons as to why some Business Plans fail. (Ten reasons)	10	3	3
	b	Justify and elaborate using relevant real-world examples of successful business model innovation.	06	4	4
		OR			
10	a	Discuss the steps on procedure for Setting up an enterprise in Indian context.	08	4	3
	b	Illustrate the different types of business models with examples.	08	3	4

18HSI51

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Apr 2024

COMMON TO EE / CS / ET / IS / BT / EC

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

			M	CO	BT
1	1.1	The broadcasting organizations like TV, Radio have a term of copyright for _____.	01	1	2
	1.2	If a person invents a new process for recording music, he or she will likely to apply for a _____.	01	1	2
	1.3	An invention has to be _____ and _____ in order to patentable.	02	2	2
	1.4	_____ is the period of limitation for filing suit for infringement of trademark in India.	01	1	2
	1.5	If the term of the years of the assignment of patents is not mentioned, then it is assigned for _____ years.	01	2	1
	1.6	Symbol of Maharaja of Air India is a _____.	01	1	2
	1.7	The trademark is registered and can be renewed every time for a period of _____.	01	2	2
	1.8	WIPO stands for _____.	01	1	2
	1.9	Does inventions relating to atomic energy are patentable or not?	01	2	1
	1.10	Software is protected as intellectual property in India primarily under the law of _____.	01	1	2
	1.11	Validity of patent is _____ years from the date of registration.	01	2	1
	1.12	The Information Technology came into effect in the year _____.	01	3	2
	1.13	_____ firm is associate the most with Design Thinking.	01	2	2
	1.14	Design Thinking typically helps in _____.	01	3	2
	1.15	In order to register Industrial Design, design should be new and _____.	01	3	2
	1.16	Whether the shape and configuration of spring can be registered as Industrial Design in India? Analyze.	02	4	2
	1.17	_____ is an important source of idea generation due to their familiarity with the needs of market.	02	3	2

PART-B

2	a	List and explain different types of “Intellectual property rights”.	08	1	2
	b	Explain the essential features of granting patent under the Indian Patent Act.	08	2	3

3	a	What cannot be patented under the Indian Patent Act?	08	1	2
	b	Differentiate between patent and trade secrets.	08	2	2
		OR			
4	a	Identify the grounds for refusal of registration of a Trademark in India. What kind of Trademarks can be registered under the (Indian) Trade Marks Act, 1999?	08	2	2
	b	Discuss the infringement of Trademark with two decided case studies.	08	2	2
5	a	"Copyright protects the expression, not the ideas". Explain this statement with suitable illustrations.	08	2	3
	b	Explain the essential features of infringement of Copyright with two decided case studies.	08	2	2
		OR			
6	a	Define the term Industrial Design and explain the essential features of Industrial Design.	08	2	2
	b	Define Cyber Crimes. Explain the commonly categorized types of Cyber Crimes.	08	1	2
7	a	What do you understand by concept of entrepreneurship? Explain the characteristics of a successful entrepreneur. What is design thinking? Describe the design thinking process.	08	1	2
	b	Identify your own entrepreneurship style based on your personality traits, strengths and weaknesses.	08	2	3
8	a	Explain the method of Customer Centric selling skill in detail.	08	2	2
	b	Identify and explain the central ideas and principles of design thinking.	08	2	3

18HSI51

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations June 2023 (Makeup)

COMMON TO EE / CS / ET / IS / BT / EC

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Does Trademarks assure quality of the product?	01
	1.2	Traditional Knowledge is limited to any specific field of knowledge. True/False	01
	1.3	The broadcasting organizations like *TV*, Radio have a term of copyright for _____.	01
	1.4	A method of growing plants on roof top is patentable or not.	01
	1.5	If a company develops a new technology that improves its main product, what type of intellectual property can they use to stop others from copying their invention?	01
	1.6	If you file provisional specification, the complete specification is required to be filed within _____ months.	01
	1.7	_____ is a symbol generally used to identify a particular product, which indicates its source.	01
	1.8	Section 66*A* of *IT* Act, 2000 has been struck down by Supreme Court on 24th March, 2015 in _____ case.	01
	1.9	Whether the shape and configuration of spring can be registered as Industrial Design in India? Analyze	02
	1.10	If an employee learns a better way of auditing the expenses of a company from his previous employer, which he uses for auditing in the present company, then is it infringement of trade secret?	02
	1.11	A Biology Teacher in a College collects the articles written by researchers on cloning and circulates the same amongst his students. Is he guilty of infringement?	02
	1.12	Mr.Binu gets a message to share the credit card details, passwords by masquerading as trustworthy source is called as _____.	01
	1.13	Customer delight experiences frequently make their way onto social media through _____ means.	01
	1.14	List any two myths related to entrepreneurs.	02
	1.15	List two sales skill to become as effective entrepreneur.	02

PART-B

2	a	Can any invention be patented after publication or display in the public exhibition? Does the Patent Office keep information of the invention secret?	08
	b	Does the Patent Office help to select a patent attorney or agent to make patent search or to prepare and prosecute patent application? Is it mandatory to obtain prior permission from the Patent Office to file application for patent outside India or abroad?	08
3	a	List the difference between Registered and Unregistered Trademarks. What are the benefits of Trademark Registration?	08
	b	Explain the absolute grounds for refusal of registration of Trade mark.	08
		OR	
4	a	Define Trademark. Explain the method of registering trademark under the Act.	08
	b	Explain the essential features of infringement of Trade Mark with two cases.	08
5	a	Describe the Provisions of Information Technology Act, 2000 dealing with Cyber crime.	08
	b	Does plagiarism in research manuscript amount to copyright infringement? Evaluate the statement in the context of academic work. Also deliberate on the exception to Copyright infringement and Remedies for Copyright infringement in Indian Context.	08
		OR	
6	a	Copy right protects the expression, not the ideas. Justify this statement with suitable illustrations.	08
	b	Is registration compulsory for obtaining design protection? What designs are not eligible for registration under the Designs Act, 2000?	08
7	a	Evaluate the theory of David McClelland's with respect to the characteristics trait and challenges of an Entrepreneurship.	08
	b	"Entrepreneurs require high levels of creativity in all the processes they conduct". In the light of the above statement, bring out the relevance of design thinking approach in encouraging critical thinking and fostering innovation for successful and sustainable entrepreneurial activity.	08
8	a	What are the limitations of personal selling? Discuss the method of customer centric selling skill with suitable examples.	08
	b	"The reward is in the risk." "If there is no risk, there is no reward." "Life is either a daring adventure or nothing at all." All these statements are telling about the risk. Enumerate on "Risk taking as a positive trait in entrepreneurial context."	08

18HSI51

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Mar/Apr 2023

COMMON TO EE / CS / ET / IS / BT / EC

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	On what basis, the total royalty on the book sold is computed under the copyright agreement?	01
	1.2	If a person uses someone's song as a background in his movie, then such an act could be an example for _____.	01
	1.3	International copyright is governed by the _____ Convention.	01
	1.4	A business arrangement where one party allows another party to use a business name and sell its products or services is known as _____.	01
	1.5	List the characteristics of an Entrepreneur?	02
	1.6	Why are small businesses important to country's economy?	02
	1.7	From a college computer, person '*A*' learned how to copy programs. Person '*B*' asked '*A*' to copy a program for his home use. What would be the most ethical response?	02
	1.8	What are the knowledge requirements of an entrepreneur?	02
	1.9	What should be the main worry of a company's founder who asks for capital in exchange for equity shares in their venture?	02
	1.10	List the motivations to be an entrepreneur.(any two)	02
	1.11	What are: a) Manufacturing Enterprises b) Service Enterprises	02
	1.12	List the characteristics of a good trade mark.	02

PART-B

2	a	Intellectual property can be categorized into artistic or commercial as per the nature of work. Justify.	08
	b	Turmeric is a well-known remedy for various injuries and beauty treatments. State the nature of *IPR* protection given by law.	08
3	a	Briefly explain the procedure of registering a trademark.	08
	b	Briefly explain the rights conferred to the person after the trademark registration.	08
		OR	

4	a	Briefly explain the chances of refusal of a trademark application.	08
	b	Briefly explain the exceptions to the trademark rights granted to a person.	08
5	a	Briefly explain the need for protecting an industrial design.	08
	b	Briefly explain the procedure of protecting computer software.	08
		OR	
6	a	Consider you are into a media business and planning to start a new broadcasting unit. Discuss about the rights you may exercise.	08
	b	Briefly explain the constraints of the design that cannot be protected by Industrial Design rights.	08
7	a	Entrepreneurial talents are the backbone of Indian economy. Justify.	08
	b	Briefly explain how an entrepreneur perceives opportunity, plan, organize resources and succeed the goal set.	08
8	a	Briefly explain any one success and a failure story of an entrepreneur in India.	08
	b	Briefly explain how design thinking helps the entrepreneur's success.	08

16HSI51

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations April-2023

COMMON SUBJECT

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	A scientist gets an idea on how to make biodegradable plastics. How can this idea be patented by the scientist?	02
	1.2	The patent Act 1970 does not convey any monopoly right to the Inventors Assignee if the assignee is inventors close relative. Comment.	02
	1.3	Can a person register his name as a Trademark? Why?	02
	1.4	The value of the trademark depends upon the _____ associated with it.	01
	1.5	_____ is the period of limitations for filing suit for infringement of trademark in India.	01
	1.6	List any two characteristics of effective entrepreneurs.	02
	1.7	Can opposition party leader Mehul claim copyright over his speech delivered in parliament which is written by his personal secretary ?	02
	1.8	Define industrial design.	02
	1.9	Identify any two entrepreneurial myths.	02
	1.10	Jack, a college final year student, is from a well off business family. He has spotted on opportunity in his hometown related to the production of electricity using wind. Can he become an entrepreneur and turn this into a venture?	01
	1.11	What is the meaning of the term “Contextual enquiry” with regard to empathize phase in design thinking.	02
	1.12	Failure rate in the first year of engineering seems to be more and reduces as the students progress to higher semesters. How would you propose to empathize with the first year students in order to obtain insights into numerous problems they might have? (Any two)	01

PART-B

2	a	In order to be patentable, an invention must be worthy of a patent grant. The patent worthiness of an invention is assessed by the government through patentability requirements. Explain the essential requirements of patentability.	08

	b	Enumerate the procedure for obtaining a patent in India.	08
3	a	List the factors to be taken into consideration while decoding the question of deceptive similarity with two decided cases.	08
	b	List and explain the differences between the Trade mark and passing off.	08
		OR	
4	a	Explain the absolute grounds for refusal of registration of Trade mark under the act.	08
	b	Explain the criteria and purpose for ecolabels ? Does eco labeling affect a consumer choice?	08
5	a	Define copyright fair use. What are copyright exceptions such as fair use under the Copyright Act?	08
	b	Define cyber crime. Explain the different kinds of cybercrimes under the Information Technology Act?	08
		OR	
6	a	Explain the different ways of transferring the copyrights.	08
	b	Explain the piracy of industrial design with two cases.	08
7	a	List and explain any seven specific step in entrepreneurial journey.	08
	b	Explain the ways to make effective communication.	08
8	a	What is design thinking? Describe the design thinking process.	08
	b	Explain the steps towards customer centric selling skill in detail.	08

18HSI51

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb 2021

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is a Patent?	02
	1.2	List the criteria of patentability?	02
	1.3	Expand the following acronyms. a) $WIPO$ b) $TRIPS$	02
	1.4	When referring to copyrighted materials, what is meant by the term "Fair use"?	02
	1.5	List two examples each of companies which have letter mark and symbol mark as their trademark.	02
	1.6	Identify any two tools to protect Trade secret.	02
	1.7	$AICTE$ has mandated that you earn 100 activity points through community service as part of partial fulfillment of your $B.E$ degree Award for batches that started from $2018-19$. You want to work on an assignment that impacts people in a genuine way. So, you ask your mentor and some of your close friends about what can be done? One of them suggest that, there are lots of "old age homes" (Vriddhashram) where elderly people find shelter. Noticeable characteristics are that they seem to be sad most of the time with drooping shoulders. Identify any two niche ideas that come to you off the top of your mind that can bring smiles on their faces.	02
	1.8	For one of your assignment on design thinking, you are working with kids having hearing disability. Identify any two activities you can plan to empathize or immerse yourself in order to get a better understanding of what they go through on a normal day before you can design a useful solution for them.	02
	1.9	Caitlin is creating a new product for university students. She takes a design –thinking approach. Her first step is addressing who she is creating the product for and conducts research on understanding this target market. What is this step called in the design thinking process?	01
	1.10	What is the meaning of the term "Contextual enquiry" with regard to emphatize phase in design thinking?	01

	1.11	Design thinking is sometimes visualized as a linear process, but it is actually :_____.	01
	1.12	Jack, a college final year student, is from a well –off business family. He has spotted an opportunity in his hometown related to the production of electricity using wind. Can he become an entrepreneur and turn this into a venture?	01

PART-B

2	a	Describe the advantages of trade secrets compared to patenting?	10
	b	What is patent information? Briefly explain the significance of using the patent information. How does a patent document help in $R\,\&\,D$.	06
3	a	What is Trade Mark? What are the essential features of trade mark?	06
	b	Explain Infringement of Trademark with two case studies.	10
		OR	
4	a	Explain the absolute grounds for refusal of registration of Trademark.	08
	b	PayPal has accused PayTM of copying its logo to grow its own user base. The US based e-wallet, a pioneer in digital transactions, claims that PayTM had no other reason to use the same two-tone colour scheme in trademark infringement complaint with the Indian Trademark office. As it comes at a time when PayTM is enjoying massive gains from the demonetization of large notes, the complaint does seem rather opportunistic; however this is not what court will be analyzing. Though the colours of a logo may not mean anything to the common observer, smaller matters have been at the centre of trademark infringement cases in the past. By understanding the above case, explain deceptive similarity and the conditions on which it will be decided using similar case examples.	08
5	a	Distinguish between Industrial Design and Copyright.	05
	b	Define Cyber-crime. Explain the different kinds of cyber –crimes.	06
	c	Explain the rights conferred by copyrights.	05
		OR	
6	a	Explain the procedure to obtain Design rights.	08
	b	You happen to interact with a junior from your program in your college who has the required skills to be copywriter and industrial designer. He happens to be part of $DEBSOC, CARV$ and takes keen interest in contributing to production design in kannada sangha events. i) Explain him regarding the career choice he can pursue with these skills. ii) Comment on the skills you noticed which are required to excel in the above mentioned field.	08
7	a	Explain the ways to make effective communication.	08
	b	Describe the common characteristics of entrepreneurs.	08

8	a	Select a theme of your choice from the following and use design thinking principles/steps to solve. Emphasis should be on process of building empathy in potential users, creating a needs statement, brainstorming and ideation and generating concepts. (prototype building and testing the ideas is not included in the scope). (your thought process is being examined) i) Portable drinking water in public places and reduce *PET* bottles. ii) Reduction of food waste in lavish Indian weddings and other programs. iii) Wet and dry waste segregation for effective recycling and disposal.	06
	b	Customer is an unpaid employee as in case of an *ATM* vestibule. Position yourself in year 2008 and look forward to improve the facilities that would enable you to provide better services with respect to safety, smooth transaction by understanding the needs of the customer.	05
	c	Take a topic or a problem over which you want to brainstorm with your group of 4 people. Quote the problem definition. What would be your inputs to the problem genesis and solution space? Summarize.	05

16HSI51

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb-2021

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	If a patent application is filed but a patent has not yet been issued, the applicant usually places the words patent _____ on the article.	01
	1.2	The royalty should not exceed _____ % of the net-ex-factory sale price in bulk of the patented article in respect of medicine or food.	01
	1.3	If you file provisional specification, the complete specification is required to be filed within _____ months.	01
	1.4	The term of an Indian patent is _____ years from the date of filing in respect of food and drugs.	01
	1.5	_____ is a product formula, pattern, design, and compilation of data, customer list or other business secret.	01
	1.6	Hall mark is an example for _____ mark.	01
	1.7	For licensing of Trademarks, the application for recordal of the registered user in India should be made within _____ months from the date of the agreement.	01
	1.8	An assignment where the owner transfers the rights and value of the trademark as associated with the product it sells, is known as _____.	01
	1.9	Through what mode does the central government can acquire a patent from the patentee for public purpose?	01
	1.10	If you write an original story, what type of intellectual property gives you the right to decide who can make and sell copies of your work?	01
	1.11	As per Design Thinking, empathy means _____.	01
	1.12	Finding fresh and creative solutions to problems is known as _____.	01
	1.13	In the concept of design thinking, two common categories of prototypes used by designers are _____ and _____.	02
	1.14	List any two myths about entrepreneurship.	02
	1.15	Compare Patent and Copyright.	02
	1.16	Differentiate between Copyrights and Industrial designs.	02

PART-B

2	a	Explain different methods of transferring the Patent Rights.	08
	b	Classify and explain the Intellectual Property Rights with suitable examples.	08
3	a	Differentiate between Infringement and Passing off.	05
	b	List the factors to be considered for deceptive similarity in infringement of Trademark.	03
	c	Explain the different modes of transferring the Trademark.	08
		OR	
4	a	Identify and explain the grounds for refusal of registration of a trademark in India.	08
	b	A British manufacturer of biscuits has been selling cookies in Malaysia for more than 20 *years* and has had a registered trademark 'ChipsMore' for these goods during that time. Two years ago, a Malaysian company started manufacturing and selling cookies under the brand "ChipsPlus'. Analyze the case study and comment on Infringement and Passing of.	08
5	a	List and explain the essential characteristics of a design.	04
	b	Describe the process for registration of the Industrial design.	06
	c	Explain the acts which do not constitute Infringement of Copyright.	06
		OR	
6	a	List and explain different types of Cyber-crimes.	08
	b	List and explain the rights conferred by Copyrights.	04
	c	List and explain the rights conferred by Broadcasting Reproduction Rights.	04
7	a	Identify and explain common characteristics of successful Entrepreneurs.	08
	b	List and explain any four strengths and weaknesses of an Entrepreneur.	08
8	a	Explain the first three steps of Design Thinking process with a suitable example.	10
	b	Why people are interested in becoming Entrepreneurs? Give reasons.	06

16HSI51 / 12HSI61

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Nov/Dec-19

Common CS / EE / EC / IS / TE / AS

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	If a person invents a new process for recording music, he or she will likely apply for a _____.	01
	1.2	Intellectual Property Rights (*IPR*) protect the use of information and ideas that are of _____.	01
	1.3	If somebody wants to develop a new method of horticulture, can he get the patent for the same?	02
	1.4	Trade-secret means _____.	01
	1.5	An invention has to be _____ and _____ in order to be patentable.	02
	1.6	Hall Mark is _____ mark.	01
	1.7	The value of the trade mark depends upon the _____ associated with it.	01
	1.8	_____ is the period of limitation for filing suit for infringement of trademark in India.	01
	1.9	_____ is an international treaty administered by *WIPO* to facilitate international trademark registration.	01
	1.10	Plan of a building can be protected by _____	01
	1.11	If the term of the years of the creative work assignment is not mentioned, then it is assigned for _____ years	01
	1.12	If an employee in a company develops a program, would this employee own the copyright?	02
	1.13	Compulsory license will be granted only after the expiry of _____ years.	01
	1.14	In a successful copyright infringement case, the plaintiff can recover _____	01
	1.15	The word entrepreneur has been taken from the _____ language.	01
	1.16	List two sales skill necessary to become an effective entrepreneur.	01
	1.17	Why should an entrepreneur do a feasibility study for starting a new venture?	01

PART-B

2	a	What is Intellectual Property? Explain different types of Intellectual property.	08
	b	Describe the procedure for obtaining a patent.	08
3	a	Explain the factors to be considered for deceptive similarity in infringement of Trade Mark.	08
	b	Define Trade Mark? Explain the features of registrable and non-registrable Trademark.	08
		OR	
4	a	Discuss the functions and features of Trade Mark with illustrations.	08
	b	Explain the infringement of Trademark with two decided case studies.	08
5	a	What is the object of registration designs? Mention the types of designs not registrable under the Act?	08
	b	"The philosophical justification for including computer programmes within the definition of 'literary work' has been that computer programmes are also products of intellectual skill like any other Literary work". Discuss	08
		OR	
6	a	Explain the rights of performer and broad casting organization under law of copyright.	08
	b	Can copyright be assigned? If so, what is the mode of assigning copyright?	08
7	a	What do you understand by concept of entrepreneurship? Explain the characteristics of a successful entrepreneur.	08
	b	List the six myths about Entrepreneurship. Write a note on any one of the successful entrepreneurs.	08
8	a	List and explain the characteristics of different entrepreneurial styles.	08
	b	What is design Thinking? Describe the design thinking process.	08

16HSI51 / 16HSI61 / 12HSI51 / 12HSI61

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V / VI Semester B. E. Fasttrack Examinations Jul-19

Common to all branches

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	X made minor improvements over his patent expired drug which is used to cure lucemia or cancer disease. Can he/she is eligible to hold fresh patent on the drug with minor improvement to make the drug easy to swallow.	02
	1.2	Can a manufacturer register the trademark 'Comfort' for his footwear?	01
	1.3	Give the reason why generic word cannot be allowed to register as a trademark.	01
	1.4	In India _____ is issued if the price of the invention (patented article) is not affordable to common man.	01
	1.5	How do you protect the computer software which has functional relationship with the computer in India?	01
	1.6	Can employer allow to make application for patent over intellectual effort of his employee?	01
	1.7	Can a design for chocolate easter egg consisting of an inner layer of white chocolate surrounded by an outer layer of dark chocolate be registrable?	01
	1.8	Mention the essential requirement of a design to be registered under Industrial Design Act.	02
	1.9	X invented UT (ultra – technology) scan machine to diagnose heardrum disease. Can he register "UT – Scanning" as a trademark for this machine?	01
	1.10	_____ is the tool for cybercrime.	01
	1.11	The term _____ is used for corporate entrepreneurship.	01
	1.12	_____ is the means by which the entrepreneur either creates new wealth producing resources or endows existing resources with enhanced potential for creating wealth.	01
	1.13	Name any two myths about entrepreneurship.	02
	1.14	_____ are the reliable source for products and services, value added enterprise.	01
	1.15	_____ will focus on environmentally friendly products from entrepreneurs.	01
	1.16	What is customer profile?	01
	1.17	Personal selling is the one method of _____ activity available to entrepreneurs.	01

PART-B

2	a	Discuss the grounds on which conferment or granting of patent can be opposed.	10
	b	Why trade secrets are kept in secrecy? How trade secrets are protected in India?	06
3	a	Can Jasmine fragrance be registered as a trademark for perfume in India?	05
	b	Discuss the characteristics of good trademark.	05
	c	Explain the procedure for protection of unregistered trademark against infringement.	06
		OR	
4	a	Define the term trademark. Explain the functions of trademark.	05
	b	Discuss the distinctiveness of trademark with illustrations.	05
	c	Discuss the characteristics of deceptive similarity. What are the factors that the courts consider while determining deceptive similarity.	06
5	a	Explain the rights which are conferred under copyright Act in India.	05
	b	Discuss the different subject matter of copyright in which copyright existed.	06
	c	Why a design need to be registered under Design Act? Give reasons.	05
		OR	
6	a	Define Cybercrime. Discuss how traditional crime is different from cybercrime.	06
	b	Discuss who is the owner of copyright in different subject matter of copyright.	10
7	a	Distinguish between Entrepreneur and Intrapreneur.	02
	b	Discuss entrepreneurial DNA needed to be possessed by the entrepreneur.	08
	c	Explain how entrepreneurship has influenced economic development of a country.	06
8	a	What is Design Thinking?	04
	b	Discuss why customers profile need to be developed as a basic element for marketing research.	06
	c	Discuss the importance of risk taking in entrepreneurship.	06

Operations Management

IM352IA | V Semester SEE Question Papers

- Feb/Mar 2026 (IM352IA)
- Feb/Mar 2025 (IM352IA)
- Feb/Mar 2024 (21IM53)
- Mar/Apr 2023 (18IM54)
- Jan/Feb 2021 (18IM54)
- Jan/Feb 2021 (16IM55)
- Fast Track Oct 2020 (16IM55)
- Nov/Dec 2019 (16IM55)
- Fast Track Jul 2019 (16IM55)

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 320247

Course Code: IM352IA

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Operations Management

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Differentiate between goods and services.	02	1	2
1.2	What is meant by productivity in operations?	02	1	2
1.3	Differentiate between job shop and flow shop.	02	3	2
1.4	Differentiate between long-term and short-term capacity decisions.	02	3	2
1.5	What is meant by exponential smoothing?	02	3	2
1.6	What is the role of forecasting in capacity planning?	02	3	2
1.7	State the concept of Theory of Constraints (TOC).	02	3	2
1.8	What is meant by load balancing in operations?	02	3	2
1.9	What is sequencing?	02	3	2
1.10	What is bottleneck scheduling?	02	3	2

Part B

Question No	Question	M	CO	BT
2a	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before trying to commercialize the device and add it to his or her existing product line, the owner wants reasonable assurance of success. Variable costs are estimated at $7 per unit produced and sold. Fixed costs are about $56,000 per year. i. If the selling price is set at $25, how many units must be produced and sold to break even? ii. Forecasted sales for the first year are 10,000 units if the price is reduced to $15. With this pricing strategy, what would be the product's total contribution to profits in the first year?	08	1	3
2b	With suitable example, Distinguish between manufacturing processes and service processes.	08	1	2
3a	Compare process structures in services and manufacturing.	08	3	4
3b	Explain long-term capacity planning. Describe capacity timing strategies and capacity sizing strategies with suitable explanations.	08	3	4
	OR			
4a	Explain timing and sizing strategies in capacity planning with diagrams.	08	3	4
4b	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer introduced variability?	08	2	2

<table>
<tr><td>5a</td><td>Explain different forecasting methods used in operations management.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>5b</td><td>The monthly demand for units manufactured by the Acme Rocket Company has been as follows:

<table><tr><td>Month</td><td>Units</td><td>Month</td><td>Units</td></tr><tr><td>May</td><td>100</td><td>September</td><td>105</td></tr><tr><td>June</td><td>80</td><td>October</td><td>110</td></tr><tr><td>July</td><td>110</td><td>November</td><td>125</td></tr><tr><td>August</td><td>115</td><td>December</td><td>120</td></tr></table>
i. Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; α = 0.2.

ii. Calculate the absolute percentage error for each month from June through December and the MAD and MAPE of forecast error as of the end of December.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>A firm uses three forecasting models: Moving Average, Exponential Smoothing, and linear regression. Explain how forecast error is calculated and used to identify the most accurate forecasting model.</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>6b</td><td>A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as 10, 11, 10, 11, 14, 15, 14 & 15 units respectively. Calculate the forecast for all the periods.

(i) If he has followed 4 month weighted moving average (with weights 0.4, 0.3, 0.2 & 0.1).

(ii) If he has followed SES with α = 0.2 and his estimated sales for January is 11.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>7a</td><td>Evaluate resource planning approaches for service providers.</td><td>08</td><td>4</td><td>5</td></tr>
<tr><td>7b</td><td>A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

<table><tr><td>Item</td><td>Client X</td><td>Client Y</td></tr><tr><td>Annual demand forecast (copies)</td><td>2,000</td><td>6,000</td></tr><tr><td>Standard processing time (hour/copy)</td><td>0.5</td><td>0.7</td></tr><tr><td>Average lot size (copies per report)</td><td>20</td><td>30</td></tr><tr><td>Standard setup time (hours)</td><td>0.25</td><td>0.40</td></tr></table></td><td>08</td><td>3</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>8a</td><td>Explain the MRP explosion process with a neat example.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>8b</td><td>Explain how the Theory of Constraints is applied to product mix decisions.</td><td>08</td><td>3</td><td>3</td></tr>
<tr><td>9a</td><td>With suitable example, differentiate between forward scheduling and backward scheduling</td><td>08</td><td>4</td><td>2</td></tr>
<tr><td>9b</td><td>Given job processing data, identify the bottleneck machine and schedule jobs to minimize idle time.

<table><tr><td>Machine</td><td>Available Time</td><td>Required Time</td></tr><tr><td>M1</td><td>40</td><td>35</td></tr><tr><td>M2</td><td>40</td><td>45</td></tr><tr><td>M3</td><td>40</td><td>30</td></tr></table></td><td>08</td><td>3</td><td>4</td></tr>
</table>

OR

10a

Consider the following single machine scheduling problem.

Job, j	Processing time, t_j	Due date, d_j
1	12	17
2	10	14
3	10	12
4	9	20
5	14	18
6	17	27

Determine the sequence which will minimize the maximum lateness (L_{max}). Also, determine the L_{max} with respect to the optimal sequence.

08 | 4 | 3

10b

Consider the following problem in single machine scheduling with independent jobs.

Job (j)	1	2	3	4	5	6	7	8
Processing time (tj)	5	12	8	10	3	15	8	6
Wj	2	1	1	2	3	4	2	3

(i) Find the optimal sequence, which will minimize the mean flow time and also obtain the corresponding minimum mean flow time.

(ii) Determine the sequence, which will minimize the weighted mean flow time. Also find the corresponding weighted mean flow time.

08 | 4 | 3

IM352IA

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/March-2025

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Distinguish between the terms 'competitive priorities' and 'competitive capabilities'.	02	2	1
	1.2	What is meant by 'nested process' in operations management?	02	2	1
	1.3	The product-process matrix brings together three elements. List them.	02	1	2
	1.4	What is meant by the concept 'economies of scale' in operations capacity planning?	02	2	2
	1.5	List any four basic patterns of demand time series.	02	1	3
	1.6	What is 'forecast error' in the context of demand forecasting?	02	2	3
	1.7	What is meant by 'bottleneck' in the context of constraint management?	02	2	3
	1.8	List the inputs required for MRP explosion.	02	1	3
	1.9	What is shortest processing time sequencing in scheduling?	02	2	4
	1.10	Expand the acronym 'CDS heuristic'.	02	1	4

PART-B

2	a	Identify nine competitive priorities used in operations strategy and explain how a consistent pattern of decisions can develop organizational capabilities.	10	2	1
	b	Describe the process view of operations in typical organizations with the help of a block diagram.	06	2	1
3	a	Explain the four inventory strategies that should be coordinated with process choice in a manufacturing environment. Provide example for each of the strategy.	10	3	2
	b	Provide a note on the three elements in a customer-contract matrix in service processes.	06	2	2
		OR			
4	a	Operations managers must examine three dimensions of capacity strategy before making capacity decisions. What are they? Explain each with suitable examples.	10	3	2
	b	Enumerate the four steps of systematic approach to long term capacity decisions.	06	2	2

5	a	Compute a 3 week moving average forecast for the arrival of medical clinical patients in week 4. The number of arrivals for the past 3weeks was as follows. $Week$ | $Patients$ 1 | 400 2 | 380 3 | 411 If the actual number of patient's arrival in week 4 is 415, what is the forecast error for week 4? Compute the forecast for week 5.	10	3	3
	b	Provide a note on the following types of error along with the formula used to compute the error. i) Mean absolute deviation ii) Mean absolute percentage error.	06	2	3
		OR			
6	a	The manager of a pizza shop must forecast the weekly demand for the special pizzas so that he can order pizza shells weekly. Recent demand has been as follows: $Week$ | $Pizzas$ | $Week$ | $Pizzas$ $June$ 2 | 50 | $June$ 23 | 56 $June$ 9 | 65 | $June$ 30 | 55 $June$ 16 | 52 | $June$ 7 | 60 Forecast the demand for pizza for June 23 to July 14 by using the simple moving average method with $n = 3$. Then repeat the forecast by using the weighted moving average method with $n = 3$ and weights of $0.50; 0.30$ and 0.20 with 0.50 applying to the most recent demand.	10	4	3
	b	"Big data characterized by 3V's holds a lot of promise for predicting the demand for products and services". Elaborate on 3V's and big data role in forecasting.	06	3	3
7	a	A plant manager received marketing's latest forecast of broadcaster roles for the next year. She wants its production line to be designed to make 2400 spreads per week atleast the next 3 months. The plant will operate 40 hours per week. i) What should be the line's cycle time? ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? iii) Suppose she finds a solution that requires only 5 stations, what would be the line's efficiency?	10	4	3
	b	List the 7 key principles of 'theory of constraints'.	06	1	3
		OR			
8	a	Create a bill of materials for a ceiling fan. Provide part number, quantity, unit measure and description for each item.	10	5	3
	b	What is meant by lot sizing rule in the context of MRP explosion? Differentiate between the fixed order quantity rule and periodic order quantity rule.	06	2	3
9	a	Consider the following single machine schedule problem. $Job\ (i)$ | 1 | 2 | 3 | 4 | 5 $Processing\ time\ (t_j)$ | 15 | 4 | 5 | 14 | 8 Find the optimal sequence which will minimize the mean flow time and also obtain the mean flow time.	10	3	4

	b	State the context in which minimizing the make span is important in single machine model of scheduling. Also mention the formula for computing the mean makespan.	06	3	4
		OR			
10	a	Consider the following 2 machines and 6 jobs flow shop scheduling problem. Using Johnson's algorithm, obtain the optimal sequence which will minimize the makespan. Also represent the makespan with a $GANTT$ chart.	16	4	4

$Job\ (i)$	$Machine\ 1$	$Machine\ 2$
1	5	4
2	2	3
3	13	14
4	10	1
5	8	9
6	12	11

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Examinations April/May-2024

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Differentiate between process and operation.	02	2	1
	1.2	List any two supply chain business processes.	02	1	1
	1.3	List any two strategies used to manage production and inventory.	02	1	1
	1.4	Highlight any two dimensions of capacity strategy before making capacity decisions.	02	1	1
	1.5	Mention any two qualitative methods of forecasting.	02	1	2
	1.6	Highlight the importance of forecast errors.	02	2	2
	1.7	List any two challenges faced during ERP implementation.	02	1	2
	1.8	Mention any two inputs used in MRP explosion diagram.	02	1	2
	1.9	List any two priority sequencing rules commonly used in practice.	02	1	1
	1.10	What is flow time in scheduling problem?	02	2	2

PART-B

2	a	Differentiate between competitive priorities and competitive capabilities with suitable examples.	08	2	1
	b	IKEA shows how important operations management is for its own success and the success of any type of organization. In this view identify some of the activities that IKEA's operations managers are involved in through highlighting some eight important aspects.	08	3	2
3	a	List and explain operations process characteristics. Also briefly discuss about implication of four characteristics of operations processes.	08	3	2
	b	How does the degree of customer contact, customization and process characteristics help in structuring processes in a service organization?	08	3	2
		OR			
4	a	Explain the concept of economies of scale when planning capacity for long term implementation.	08	2	2
	b	The best choice for a manufacturing process depends on the volume and degree of customization required of the process. Justify the above statement with the help of a suitable matrix.	08	3	3

5	a	PM Computer Services assembles customized personal computers from generic parts. Formed and operated by part-time UMass Lowell students Paulette Tyler and Maureen Becker, the company has had steady growth since it started. The company assembles computers mostly at night, using part-time students. Paulette and Maureen purchase generic computer parts in volume at a discount from a variety of sources whenever they see a good deal. Thus, they need a good forecast of demand for their computers so that they will know how many parts to purchase and stock. They have compiled demand data for the last 12 months as reported below.			

Period	Month	Demand	Period	Month	Demand
1	Jan	37	7	Jul	43
2	Feb	40	8	Aug	47
3	Mar	41	9	Sep	56
4	Apr	37	10	Oct	52
5	May	45	11	Nov	55
6	Jun	51	12	Dec	54

i. Use exponential smoothing with smoothing parameter $\alpha = 0.3$ to compute the demand forecast for January (Period 13).

ii. Use exponential smoothing with smoothing parameter $\alpha = 0.5$ to compute the demand forecast for January (Period 13). 10 3 3

b With the help of suitable examples explain any two qualitative methods of forecasting. 06 3 3

OR

6 a The Instant Paper Clip Office Supply Company sells and delivers office supplies to companies, schools, and agencies within a 50-mile radius of its warehouse. The office supply business is competitive, and the ability to deliver orders promptly is a big factor in getting new customers and maintaining old ones. (Offices typically order not when they run low on supplies, but when they completely run out. As a result, they need their orders immediately.) The manager of the company wants to be certain that enough drivers and vehicles are available to deliver orders promptly and that they have adequate inventory in stock. Therefore, the manager wants to be able to forecast the demand for deliveries during the next month. From the records of previous orders, management has accumulated the following data for the past 10 months:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Orders	120	90	100	75	110	50	75	130	110	90

a) Compute the monthly demand forecast for February through November using the naive method.

b) Compute the monthly demand forecast for April through November using a 3-month moving average.

c) Compute the monthly demand forecast for June through November using a 5-month moving average.

d) Compute the monthly demand forecast for April through November using a 3-month weighted moving average. Use weights of 0.5, 0.33, and 0.17, with the heavier weights on the more recent months. 10 3 3

b Examine the various factors affecting demand forecasting. 06 2 3

7	a	With the help of suitable examples explain seven key principles of the theory of constraints.	08	3	3
	b	What are the key challenges associated with ERP implementation? How can organizations mitigate these challenges?	08	4	4
		OR			
8	a	With the help of suitable MRP explosion diagram highlight the important inputs to MRP.	08	2	3
	b	How do you manage bottlenecks in manufacturing and service processes? Explain with the help of case example.	08	4	2
9	a	There are seven jobs, each of which has to be processed on two machines. A and B in the order AB. Processing times are given in the following table:			

Job	Machine A	Machine B
1	3	8
2	12	10
3	15	10
4	6	6
5	10	12
6	11	1
7	9	3

		Determine a sequence of these jobs that will minimize the total elapsed time. Also find total elapsed time and idle time for machines A and B.	10	4	5
	b	Give Johnson's procedure for determining an optimal sequence for processing 'n' items on two machines. Give justification of the rule used in the procedure.	06	2	4
		OR			
10	a	Consider the following 2 machines and 6 jobs flow shop scheduling problem using Johnson's algorithm. Obtain the optimal sequence which will minimize the make span. Processing times are in minutes.			

Jobs	Machine 1	Machine 2
1	5	4
2	2	3
3	13	14
4	10	1
5	8	1
6	12	11

	b	What is the no passing rule in a sequencing algorithm? Explain the principal assumptions made while dealing with sequencing problems.	06	2	3

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations March /April-2023

Industrial Engineering and Management

OPEARATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Explain the term value-added.	02
	1.2	List the nine competitive priorities.	02
	1.3	What is the objective of a process strategy?	02
	1.4	List the four process strategies.	02
	1.5	How is capacity planning measured.	02
	1.6	What are the five steps of theory of constraints?	02
	1.7	What are the causal forecasting methods?	02
	1.8	What is the purpose of a tracking signal and how it is calculated?	02
	1.9	What is the key difference between ERP and MRP?	02
	1.10	List the inputs for MPS.	02

PART-B

2	a	Explain the term operations management and Identify the three major functional areas of business organizations and briefly describe how they are interrelated?	08
	b	Describe the new trends and challenges in operations management.	08
3	a	Describe the four basic process strategies.	08
	b	List the different tools for process analysis and design and explain them.	08
		OR	
4	a	Define strategic fit. Describe the elements of strategic fit.	08
	b	What is six sigma model of quality improvement? Describe the five steps of six sigma improvement model.	08
5	a	Following data is given for a rocket propulsion: • Expansion for industry of 1 ton capacity curing furnace. • Furnace time= 30mins • System efficiency= 80% • Required output=16 tons per 8 hours • Land efficiency is estimated at 50% of system capacity. i) Determine the number of furnaces required? ii) Estimate the percentage of time the furnace will be idle.	08

	b	The Peach tree Airport in Atlanta serves light aircraft. It has a single runway and one air traffic controller to land planes. It takes on airplane 12 minutes to land and clear the runway. Planers arrive at the airport at the rate of 4 per hour. i) Determine the average number of planes that will stack up waiting to land. ii) Find the average time a plane must wait in line before it can land. iii) Calculate average time it takes a plane to clear the runway.	08
		OR	
6	a	How are bottlenecks removed in manufacturing? Explain.	08
	b	Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week. The plant will operate 40 hours per week. i) What should be the line's cycle time? ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? iii) Suppose that she finds a solution that requires only five stations, what would be the line's efficiency?	08
7	a	Given the data below, what is the simple linear regression model that can be used to predict sales in future weeks?	

$Week$	$sales$
1	150
2	157
3	162
4	166
5	177

			08
	b	What is Delphi method? Describe its main advantage and limitations.	08
8	a	Solve the following problem of 5 jobs through 2 machines A and B in the order $A-B$. the processing time of each jobs on a specific machine in hours is given in the table below.	

Job	$Machines$	
	A	B
1	6	8
2	7	3
3	12	9
4	5	15
5	10	11

		Determine the sequence which minimizes the total elapsed time and also find idle time of each machine.	08
	b	List and explain the inputs and outputs of MRP.	08

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb–21

Industrial Engineering and Management

OPEARATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Calculate the productivity for the following operations: Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week.	02
	1.2	A hospital is considering a new procedure to be offered at Rs. 200 per patient. The fixed cost per year would be Rs.1,00,000 with total variable cost of Rs.100 per patient. What is the break-even quantity for this service?	02
	1.3	State three approaches to inventory that should be coordinated with process choice.	02
	1.4	List the five techniques that are effective for documenting and evaluating processes.	02
	1.5	Write the formula to calculate utilization of any equipment.	02
	1.6	Differentiate between backlog and back order.	02
	1.7	Give one example each for Dependent and Independent demand.	02
	1.8	Define Line balancing.	02
	1.9	Define Value Stream Mapping (VSM). Write any 4 symbols used to draw VSM.	02
	1.10	Differentiate between push and pull workflow.	02

PART-B

2	a	Consider amazon.com, whose Website enjoys millions of “hits” each day and puts customers in touch with millions of services and products. What are amazon.com’s competitive priorities and what should its operations strategy focus on?	06

	b	A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6 respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If the small facility is built and demand is low, there is no reason to expand and the payoff is $200,000.If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate maximum demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7 respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative- building a small facility or building a large facility-has the higher expected payoff?	10
3	a	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability?	06
	b	A work sampling study was conducted to determine the proportion of the time a worker is idle. The following information was gathered on a random basis.	

Day	Number of times worker idle	Total number of observations
Monday	17	44
Tuesday	18	56
Wednesday	14	48
Thursday	16	60

i) Based on these preliminary results, what percent of the time is the worker working?

ii) If the idle time is judged to be excessive, what additional categories might you add to a follow-up work sampling study to identify the root causes? 10

OR

4 a Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings.

i) Airport
ii) Bank
iii) Classroom
iv) Product designers' office 08

b Vera Johnson and Merris Williams manufacture vanishing cream. Their packaging process has four steps: (1) mix, (2) fill, (3) cap, and (4) label. They have had the reported process failures analyzed, which shows the following:

Process Failure		Frequency
Lumps of unmixed product		7
Over or under filled jars		18
Jar lids did not seal		6
Labels rumpled or missing		29
	Total	60

08

5	a	Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for atleast the next three months. The plant will operate 40 hours per week. i) What should be the line's cycle time? ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time?	08
	b	A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Base on the following table of information; determine how many machines are needed at the copy center.	

Items	Client X	Client Y
Annual demand forecast (copies)	2000	6000
Standard processing time (hr/copy)	0.5	0.7
Average lot size (copies/report)	20	30
Standard setup time (hours)	0.25	0.4

08

OR

6	a	Take a process that you encounter on a daily basis, such as the lunch cafeteria or the journey from your home to school/ work, and identify the bottlenecks that limit the throughput of this process.	06
	b	The Dahlia Medical Center has 30 labour rooms, 15 combination labour and delivery rooms, 3 delivery rooms, and 1 special delivery room reserved for complicated births. All these facilities operate around the clock. Time spent in labour room varies from hours to days, with an average of about a day. The average uncomplicated delivery requires about 1 hour in a delivery room. During an exceptionally busy 3-day period, 109 healthy babies were born at Dahlia Medical center. Sixty babies were born in separate labour and delivery rooms, 45 were born in combined labour and delivery rooms, and only 4 babies required a labour room and the complicated-delivery room. Which of the facilities (labour rooms, combination labour and delivery rooms, or delivery rooms) had the greatest utilization rate?	10
7	a	The monthly demand for units manufactured by the Acme Rocket Company has been as follows:	

Month	May	June	July	Aug	Sep	Oct	Nov	Dec
Units	100	80	110	115	105	110	125	120

		Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; a = 0.2.	10
	b	Consider the master flight schedule of a major airline, such as Air New Zealand. Discuss the ways in which it is analogous to a master production schedule for a manufacturer.	06

8	a	Explain the eight types of wastes.	06
	b	Consider the following 3 machines and 5 jobs flow shop problem.	

Job, i	Processing Time		
	Machine 1	Machine 2	Machine3
1	10	10	12
2	12	8	20
3	16	6	14
4	12	4	20
5	20	8	8

i) Check whether the Johnson's algorithm can be extended to this problem.

ii) If yes, find the optimal sequence and the corresponding makespan. 10

16IM55

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Differentiate between internal and external customer with respect to process view.	02
	1.2	What are the four core processes nested in Operations Management?	02
	1.3	Identify four basic process decisions in Operations Management?	02
	1.4	What are two different but complementary philosophies for process design?	02
	1.5	List the measures of capacity utilization.	02
	1.6	List the steps of a Systematic Approach to Long-Term Capacity Decisions in manufacturing plants.	02
	1.7	Differentiate between constraint and bottleneck.	02
	1.8	What is Tracking Signal in forecasting? Write the formula for Tracking Signal.	02
	1.9	What are Inventory Reduction Tactics or levers?	02
	1.10	What is aggregation? List the three dimensions that companies perform aggregation.	02

PART-B

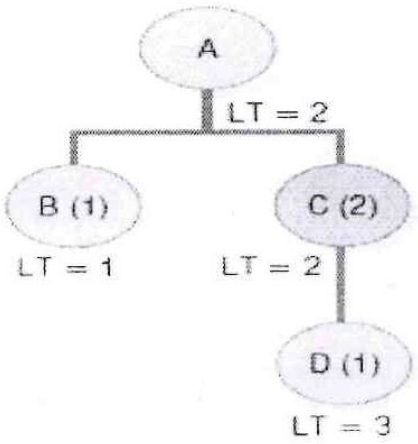
2	a	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and the responses the firm will make to changes in the environment. Identify and discuss the factors Connection between Corporate strategy and key operations management decisions with block diagram.	08
	b	Calculate the productivity for the following operations: a) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. b) A team of workers makes 400 units of a product, which is sold in the market for $10 each. The accounting department reports that for this job the actual costs are $400 for labor, $1,000 for materials, and $300 for overhead.	08

3	a	Identify and discuss with block diagram the important four decisions that are essential for an effective process design.	08
	b	Identify and discuss the strategies used for structuring the process in Manufacturing.	08
		OR	
4	a	The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit.	08
	b	Identify and discuss the key elements of process reengineering.	08
5	a	Operation managers must examine three dimensions of capacity strategy before making capacity Decisions. Identify and discuss those three dimensions of capacity strategy before making capacity decisions.	08
	b	A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.	08

Item	Client X	Client Y
Annual demand forecast (copies)	2000	6000
Standard processing time (hour/copy)	0.5	0.7
Average lot size (copies per report)	30	30
Standard setup time (hours)	0.25	0.40

		OR	
6	a	What is Theory of Constraints (TOC)? Identify seven key principles of the Theory of Constraints.	08
	b	Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week. a) What should be the production line's cycle time? b) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? c) Suppose that she finds a solution that requires only five stations, what would be the line's efficiency?	08

7	a	The forecast process itself, typically done on a monthly basis, consists of structured steps. These steps often are facilitated by someone who might be called a demand manager, forecast analyst, or demand / supply planner. However, many other people are typically involved before the plan for the month is authorized. Identify and discuss those structured steps for forecasting process.	08
	b	A plant makes monthly shipments of electric drills to a wholesaler in average lot sizes of 280 drills. The wholesaler's average demand is 70 drills a week, and the lead time from the plant is 3 weeks. The wholesaler must pay for the inventory from the moment the plant makes a shipment. If the wholesaler is willing to increase its purchase quantity to 350 units, the plant will give priority to the wholesaler and guarantee a lead time of only 2 weeks. What is the effect on the wholesaler's cycle and pipeline inventories?	08
8	a	Briefly discuss the different types of Operations Planning and Scheduling across the Organization.	08
	b	The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule: 100 units in week 2; 200 unit in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the net 8 week for items B, C, and D. The BOM for A is shown in figure and data from the inventory records are shown in table.	08



Inventory Recorded Data			
	Items		
Data Category	B	C	D
Lot Sizing Rule	POQ(P=3)	L4L	FOQ=500 units
Lead Time	1 week	2 weeks	3 weeks
Scheduled Receipts	None	200 (week 1)	None
Beginning Inventory (On Hand)	20	0	425

16IM55 / 12IM62

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Define an operations strategy	02
	1.2	Diagrammatically represent the integration between the three main functional areas of business, namely operations, finance and marketing.	02
	1.3	What are the four major process related decisions pertaining to operations management?(List them)	02
	1.4	List any four aspects of continuous flow processes.	02
	1.5	Define the terms capacity and utilization.	02
	1.6	List any four of the seven key principles of the theory of constraints.	02
	1.7	Give the precise definition of ordering cost and setup cost.	02
	1.8	Identify and list the five basic patterns of most demand time series.	02
	1.9	Clearly define the sales and operations plan(S & OP)	02
	1.10	What is the meaning of MRP explosion?	02

PART-B

2	a	Define operations management. Explain with the help of a block diagram the process view of operations management and write how processes work.	06
	b	A manager is trying to decide whether to build a small, medium or large facility. Demand can be low, average or high with estimated probabilities being $0.25, 0.40$ and 0.35 respectively. A small facility is expected to earn an after tax net present value of just \$18,000 if demand is low. If demand is average, the small facility is expected to earn \$ 75,000; It can be increased to medium size to earn a net present value of \$60,000.If demand is high, the small facility is expected to earn \$75,000 and can be expanded to medium size to earn \$60,000 or to large size to earn \$ 125,000 and medium sized facility is expected to lose an estimated \$25,000 if demand is low and earn \$140,000 if demand is average. If the demand is high the medium sized facility is expected to earn a net present value of \$150,000; It can be expanded to a large size for a net pay off of \$145,000.	

		If a large facility is built and demand is high earnings are expected to be \$220,000.If demand is average for the large facility the present value is expected to be \$125,000; if demand is low, the facility is expected to lose \$60,000. i) Draw a decision tree for this problem? ii) What should the management do to achieve the highest expected pay off?	10
3	a	Explain why processes exist everywhere in all organizations. Take the following supply chain activities and give a brief description for each. i) Warehousing ii) Customer service iii) Logistics.	08
	b	Briefly discuss on the process strategy decisions. Use a block diagrammatic representation to bring out the connections among major process strategy decisions for effective processes.	08
		OR	
4	a	Define the following terms clearly: i) Customer increment ii) Resource flexibility iii) Capital intensity iv) Economics of scope.	08
	b	Position a process on the customer-contact matrix or product process matrix. Bring out the characteristics of the following types of process choices: i) Job processes ii) Batch processes iii) Line processes iv) Continuous flow processes.	08
5	a	Outline the systematic approach to long term capacity decisions using the four steps to arrive at the capacity related decisions	08
	b	A copy center in an office building prepares band reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to make, collate and bind each copy depends on among other factors the number of pages. The center operates 250 days per year, with one 8 hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowances built in to the time standards) is best. It correctly has 3(three) copy machines. Based on the following table of information, determine how many machines are needed at the copy center.	

Item	*Client X*	*Client Y*
Annual demand forecast(Copies)	2,000	6,000
Standard processing time(Hours/copy)	0.5	0.7
Average lot size (copies per report)	20	30
Standard setup time(hours)	0.25	0.40

08

OR

6 a Johnson logs wants to setup a line to serve 60 customers per hour. The work elements and their precedence relationship are shown in the following table 6(a)

i) What is the minimum number of stations?

ii) How many stations are required using the longest work element decision rule?

iii) Suppose that a solution requiring five stations is obtained what is the efficiency?

Table 6(a)-Johnson logs assembly line data

Work element	*Time(Sec)*	*Immediate predecessor*
A	40	*None*
B	30	A
C	50	A
D	40	B
E	6	B
F	25	C
G	15	C
H	20	D, E
I	18	F, G
J	30	H, I
Total	274	

08

b Clearly outline the central ideas enshrined in the theory of constraints. List the key principles of the theory of constraints. List any two benefits of applying TOC to practical scenarios. 08

7 a The monthly demand for units manufactured by the $ACME$ company has been as follows:

Month	*Units*	*Month*	*Units*
May	100	*September*	105
June	80	*October*	110
July	110	*November*	125
August	115	*December*	120

i) Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units $\alpha = 0.2$

ii) Calculate the absolute percentage error for each month from, June through December and also the MAD and $MAPE$ of forecast error as of the end of December. 08

b Nelson's hardware store stocks a 19.2 volt cordless drill that is a popular seller. Annual demand is 5000 units. The ordering cost is \$15 and the inventory holding cost is \$4/unit/year.

i) What is the economic order quantity?

ii) What is the total annual cost for this inventory item? 08

8	a	Clearly bring out the distinction between the two basic planning strategies of chase strategy and level strategy.	04
	b	A company manufactures wooden decors. Management schedules overtime every week end to reduce the back logs on most popular models. The automatic routing machine is used to cut certain types of edges on the desktops. The following orders need to be scheduled for the routing machine.	

$Order$	$Time\ since\ order\ arrived(hours\ ago)$	$Estimated\ machine\ time(hours)$	$Due\ date\ (hours\ from\ now)$
1	1	9	20
2	0	7	21
3	6	8	12
4	5	3	8
5	3	12	18

The due date reflect the needs for the order to be at its next operation

i) Develop separate schedules by using $FCFS$ and EDD rules. Compare the schedules on the basis of average flow time and average past due hours.

ii) Comment on the performance of these two rules relative to these measures. 08

	c	Briefly explain the concepts related to MRP explosion with an example	04

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Differentiate between internal and external customer with respect to process view.	02
	1.2	What are the four core processes nested in Operations Management?	02
	1.3	Identify four basic process decisions in Operations Management?	02
	1.4	What are two different but complementary philosophies for process design?	02
	1.5	List the measures of capacity utilization.	02
	1.6	List the steps of Systematic approach to Long-Term capacity Decisions related to manufacturing plants.	02
	1.7	Differentiate between constraint and bottleneck.	02
	1.8	What is Tracking signal in forecasting? Write the formula for tracking signal.	02
	1.9	What are Inventory reduction tactics or levers?	02
	1.10	What is aggregation? List the three dimensions that companies perform aggregation.	02

PART-B

2	a	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and the responses the firm will make to changes in the environment. Identify and discuss the factors connection between corporate strategy and key operations management decisions with block diagram.	08
	b	Calculate the productivity for the following operations: i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. ii) A team of workers makes 400 units of a product, which is sold in the market for \$10 each. The accounting department reports that for this job the actual costs are \$400 for labour, \$1000 for materials and \$300 for overhead.	08
3	a	Identify and discuss with block diagram the important four decisions that are best understood as an effective process design.	08
	b	Identify and discuss the strategies used for structuring the process in manufacturing.	08

OR

4 a The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective.
As an industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit. 08

b Identify and discuss the key elements of process reengineering. 08

5 a Operations managers must examine three dimensions of capacity strategy before making capacity decisions.
Identify and discuss those three dimensions of capacity strategy before making capacity decisions. 08

b A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15%(beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

$Item$	$Client\ X$	$Client\ Y$
$Annual\ demand\ forecast(copies)$	2,000	6,000
$Standard\ processing\ time(hour/copy)$	0.5	0.7
$Average\ lot\ size(copies\ per\ report)$	20	30
$Standard\ setup\ time(hours)$	0.25	0.40

08

OR

6 a What is Theory of Constraints (TOC)? Identify seven key principles of the Theory of Constraints 08

b Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week.

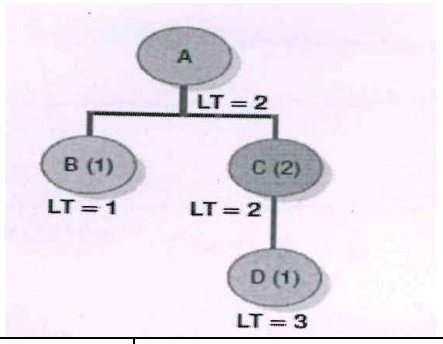
i) What should be the line's cycle time?

ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time?

iii) Suppose that she finds a solution that requires only five stations. What would be the line's efficiency? 08

7 a The forecast process itself, typically done on a monthly basis, consists of structured steps. These steps often are facilitated by someone who might be called a demand manger, forecast analyst, or demand / supply planner. However, many other people are typically involved before the plan for the month is authorized. Identify and discuss those structured steps for forecasting process. 08

	b	A plant makes monthly shipments of electric drills to a wholesaler in average lot sizes of 280 drills. The wholesaler's average demand is 70 drills a week, and the lead time from the plant is 3 weeks. The wholesaler must pay for the inventory from the moment the plant makes a shipment. If the wholesaler is willing to increase its purchase quantity to 350 units, the plant will give priority to the wholesaler and guarantee a lead time of only 2 weeks. What is the effect on the wholesaler's cycle and pipeline inventories?	08
8	a	Briefly discuss the different types of Operation Planning and scheduling across the Organization.	08
	b	The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule:100 units in week 2; 200 units in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the next 8 weeks for items B, C and D. The BOM for A is shown in Figure and data from the inventory records are shown in Table.	



	Item		
Data category	B	C	D
$Lot\ sizing\ rule$	$POQ(P=3)$	$L4L$	$FoQ = 500 units$
$Lead\ time(LT)$	$1\ week$	$2\ weeks$	$3\ weeks$
$Scheduled\ receipts$	$None$	$200(week\ 1)$	$None$
$Beginning\ (on\ hand)\ inventory$	20	0	425

08

16IM55/12IM62

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU)

V Semester B. E.Fast Track ExaminationsJuly-19

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Diagrammatically represent the interactions between different functional areas of business-mainly marketing, operations and finance.	02
	1.2	List out any four of the key differences between manufacturing and service processes.	02
	1.3	List out the four major process strategy decisions.	02
	1.4	Bring out any four distinctions between job process and batch process.	02
	1.5	List out any two of the long term capacity and short term constraint management capacity decisions as part of overall capacity management.	02
	1.6	Articulate unambiguously and cleanly any four of the seven key principles of the theory of constraints.	02
	1.7	Write down the formula for safety stock with the assumptions that demand distribution and lead time distribution are measured in the same units. Demand and lead time are independent. Explain the notations used in the formula.	02
	1.8	List out any four of the judgment methods for demand forecasting with a one line description for each of these.	02
	1.9	Define the following terms clearly. i. Sales and operations plan ii. Resource plan	02
	1.10	Clearly define the periodic order quantity and lot for lot lot-sizing role to bring out its working mechanics.	02

PART-B

2	a	How does the process view and the supply chain view of operations management, help us in understanding the need and importance of operations management as a strategic function in the organization? Elaborate on these two perspectives of operations management.	08

	b	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before trying to commercialize the device and add it to his or her existing product line, the owner wants reasonable assurance of success. Variable lots are estimated at $7 per unit produced and sold. Fixed lots are about$500,000 *per year*. i) Forecasted sales for the first year are 10,000 units. If the price is reduced to$15. With this pricing strategy, what would be the products total contribution to profits in the first year? ii) If the selling price is set at $25, how many units must be produced and sold to break even? Use both algebraic and graphical approaches.	08

3	a	Construct the product-process matrix and show in this matrix process characteristic categories, level of product customization and the elements of volume. Elaborate on the type of manufacturing process structure that exist in various types of organization briefly.	08
	b	"After a process is documented, metrics data collected and disconnects identified, the process analyst or design team puts together a set of changes that will make the process better". Compare this with doctor diagnosing the patient and recommending treatments based on the diagnosis. Explain how questioning and brainstorming will assist the process designer in analyzing the gaps in the process. Take an example to support your answer.	08
		OR	
4	a	Develop the customer-contact matrix and indicate the elements of degree of customer contact, customization and process characteristic. Elaborate in the process divergence and flow based on the customer-contact matrix. How does this matrix provide the starting point for evaluating and improving a process?	08
	b	Capital intensity is the mix of equipment and human skills in the process. Carefully examine the advantages and disadvantages of automating manufacturing process. Distinguish between fixed automation and flexible automation. Give two examples for the fixed and flexible automation respectively. Justify automation in the context of capital intensity.	08

5	a	Outline the procedure for carrying out the systematic approach to long term capacity decisions. As part of the approach, discuss some of the qualitative and quantitative methods for evaluating the alternatives for capacity decisions.	08
	b	You have been asked to put together a capacity plan for a critical operation at the Surefoot Sandal company. Your capacity measure is number of machines. Three products (men's, women's and children's sandal) are manufactured. The time standards (processing and setup), lot sizes and demand forecast are given in the Table 5b below. The firm operates 8-hour shifts, 5 days per week, 50 weeks per year. Experience shows capacity cushion of 5% is sufficient.	

Table 5b: Data pertaining to products

Product	Time standards		Lot size (pairs/lot)	Demand forecast (pairs/year)
	Processing (hours/pair)	Setup (hours/pair)		
Men's sandal	0.05	0.5	240	80,000
Women's sandal	0.10	2.2	180	60,000
Children's sandal	0.02	3.8	360	120,000

i) If the operation currently has two machines, what is the capacity gap?

ii) How many machines are needed? 08

OR

6 a Elaborate on the concept known as economies of scale. Explain how the concept of economies of scale can drive cost down, when output increases. Highlight the four reasons on the cost impact of economies of scale. Discuss briefly diseconomies of scale. 08

b In order to meet holiday demand, pizza house requires a production line that is capable of producing 50 pizzas per week, while operating only 40 hours per week. There are only 4 steps required to produce a single pizza with respective processing times of 5min, 5min, 45min and 15min.

i) What share is the line's cycle time?

ii) What is the smartest number of workstations pizza house could hope for in designing the line considering this cycle time?

iii) Suppose that pizza house finds a solution that requires only four stations. What would be the efficiency of this line? 08

7 a The office supply shop estimates that the average demand for a popular ball point pen is 12,000 pens per week with a standard deviation of 3000 pens. The current inventory policy calls for a replenishment orders of 156000 pens. The average lead time from the distributor is 5weeks, with a standard deviation of 2 weeks. If management wants a 95percent cycle service level, what should the reorder point be? 05

b Nelson's hardware stocks a 19.2volt cordlessdrill that is a popular seller. Annual demand is 5000units, the ordering cost is $15 and the inventory holding cost is $4/unit/year.

i) What is the economic order quantity

ii) What is the total annual cost for this inventory item? 05

c Arrival of patients into a clinic has been noted down. The number of arrivals for the past 3 weeks is as follows.

Week	Patient arrivals
1	400
2	380
3	411

		It is now the end of week 3, so the arrival number of patients is known to be 411 patients. Using $alpha = 0.10$ ($\alpha = 0.10$) i) Calculate the exponential smoothing forecast for week 4 ii) What was the forecast error for week 4 if the actual demand turns out to be 415? iii) What is the forecast for week 5?	06

8	a	Represent in the form of block diagram, the managerial inputs that go into developing sales and operations plan, from the functional areas of operations, distribution and marketing, materials, accounting and finance, engineering and human resource. Discuss the supply options for operations planning and scheduling.	08
	b	The Mowry machine shop still has 5 jobs to be processed as of 8AM today (day 23) at its bottlenecks operations. The day when the order arrived, processing time and promised due dates are given in the following table- Table 8b. The jobs are listed in the order of arrivals.	

Table 8b

Job	Time since order arrived (days ago)	Processing time (days)	Due date (days from now)
A	13	8	36
B	15	4	42
C	17	4	39
D	22	3	53
E	24	10	45

i) Develop separate schedules by using FCFS and EDD rules. Compare the schedules on the basis of average flow time and average days past due.

ii) Which rule gives the best schedule, in your judgment? Why? 08

Finance Accounting and Costing

IM254TA | V Semester SEE Question Papers

- Feb/Mar 2026 (IM254TA)
- Feb/Mar 2025 (IM254TA)

Question Paper ID: 217618

Course Code: IM254TA

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Finance Accounting and Costing

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	List two qualitative characteristics of accounting information.	02	1	1
1.2	Comment on the Matching Concept with the help of an example.	02	1	1
1.3	Why does the single-entry system make it difficult to detect fraud?	02	1	2
1.4	Record the transaction: *Cash received from Rohan ₹5,000* in a Journal entry.	02	2	3
1.5	How does depreciation affect profit and asset value?	02	1	2
1.6	Name any two expenses recorded in the Profit & Loss Account.	02	1	2
1.7	Define cost.	01	1	1
1.8	Classify costs based on traceability.	02	1	2
1.9	What is a cost sheet?	01	4	1
1.10	Mention any two causes of material price variance.	02	4	1
1.11	List the components of standard cost.	02	4	1

Part B

Question No	Question	M	CO	BT
2a	Critically evaluate the role of accounting in promoting corporate transparency and ethical governance with an example.	10	4	5
2b	Analyze how the Going Concern Concept influences depreciation and asset valuation.	06	2	3

<table>
<tr><td>3a</td><td>From the following ledger balances, prepare a Trial Balance.

Ledger Balances: Cash in Hand ₹6,000, Cash at Bank ₹22,000, Purchases ₹45,000, Sales ₹78,000, Furniture ₹15,000, Rent ₹5,000, Capital ₹60,000</td><td>06</td><td>2</td><td>3</td></tr>
<tr><td>3b</td><td>Explain the steps to rectify the imbalance in trial balance.</td><td>10</td><td>4</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>For the following balances, prepare a trial balance.

<table>
<tr><th>Particulars (Dr)</th><th>Amount (₹)</th><th>Particulars (Cr)</th><th>Amount (₹)</th></tr>
<tr><td>Sales</td><td>1,07,800</td><td>Capital</td><td>1,00,000</td></tr>
<tr><td>Sales Returns</td><td>2,800</td><td>Discount received</td><td>1,800</td></tr>
<tr><td>Purchases</td><td>62,400</td><td>Plant & Mach.</td><td>35,000</td></tr>
<tr><td>Purchase Return</td><td>2,400</td><td>Bank overdraft</td><td>20,000</td></tr>
<tr><td>Opening Stock</td><td>15,000</td><td>Trade expenses</td><td>1,500</td></tr>
<tr><td>Wages</td><td>10,600</td><td>Furniture</td><td>5,000</td></tr>
<tr><td>Sundry Creditors</td><td>32,500</td><td>Loose Tools</td><td>3,200</td></tr>
<tr><td>Wages Outstanding</td><td>1,400</td><td>Insurance</td><td>500</td></tr>
<tr><td>Carriage outward</td><td>1,400</td><td>Drawings</td><td>3,000</td></tr>
<tr><td>Coal and Gas</td><td>2,500</td><td>Bank charges</td><td>500</td></tr>
<tr><td>Commission</td><td>1,000</td><td>Salaries</td><td>4,500</td></tr>
<tr><td>Factory Building</td><td>1,00,000</td><td>Plant depreciation</td><td>2,000</td></tr>
<tr><td>Sundry Debtors</td><td>15,000</td><td></td><td></td></tr>
</table></td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>4b</td><td>Why is ledger known as the primary book or the principal -book of accounts? Can profit of the business and its financial position be known without maintaining ledger?</td><td>08</td><td>1</td><td>3</td></tr>
<tr><td>5a</td><td>Explain the objectives and limitations of financial statements.</td><td>08</td><td>1</td><td>1</td></tr>
<tr><td>5b</td><td>Explain why Trial Balance is prepared and its limitations.</td><td>08</td><td>1</td><td>1</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>6a</td><td>Explain how depreciation affects:
i) Profit
ii) Asset Value
iii) Capital</td><td>06</td><td>4</td><td>4</td></tr>
<tr><td>6b</td><td>Prepare a Trading Account from the following particulars for the year ended 31st March 2018

Opening Stock ₹50,000

Purchases ₹1,30,000

Sales ₹3,15,000

Factory rent ₹10,000

Purchases return ₹10,000</td><td>10</td><td>3</td><td>3</td></tr>
</table>

Sales return ₹15,000

Wages ₹8,000

Carriage on purchases ₹3,500

Manufacturing Exp. ₹8,900

Custom Duty ₹2,100

Motive power ₹2,000

Dock charges ₹3,000

Coal, Gas, and Water ₹3,000

Factory Manager Salary ₹12,000

Commission on Purchase ₹8,000

Value of Closing stock ₹60,000

7	Product A is a product produced after three distinct processes. The following information is obtained from the accounts of the company for a particular period. Production overheads are incurred Rs. 800 and is recovered at 200% of direct wages. Production during the period was 100 kg. There was no opening or closing stock. Prepare Process Accounts assuming that there is no process loss. Particulars | Total Amount (Rs.) | Process I (Rs.) | Process II (Rs.) | Process III (Rs.) Direct Material | 2,200 | 1,800 | 300 | 100 Direct Labor | 400 | 100 | 200 | 100 Direct Expenses | 500 | 300 | — | 200	16	4	3
	OR						
8a	Analyze how improper classification of cost elements (Material, Labour, Overheads) can affect managerial decision-making.	08	4	4
8b	Prepare a Cost Sheet showing Cost of Sales Particulars | Amount (₹) Direct Material | 40,000 Direct Labour | 25,000 Factory OH | 60% of DL Admin OH | 10% of Factory Cost Selling OH | 5% of Cost of Production	08	4	3

9a	KBC bearing ltd is committed to supply 48,000 bearing per annum to KMR fans on a steady daily basis. It is estimated that it costs Rs 1 as inventory holding cost per bearing per month and that the set up cost per run of bearing manufacture is Rs 3200. i) Determine the optimum run size of bearing manufacture? ii)Calculate what should be the interval between two consecutive optimum runs? iii)Find out the minimum inventory cost.	12	4	4
9b	Explain the concept of Standard Costing and state its objectives.	04	4	2
	OR			
10a	Calculate cost Material Variances and price variances. Standard: 10 kg @ ₹5 Actual: 12 kg @ ₹6	08	4	3
10b	Calculate Labour Variances. Standard: 40 hrs @ ₹20 Actual: 45 hrs @ ₹22	08	4	3

IM254TA

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/March-2025

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	Define *GAAP* and mention its importance.	02	2	1
	1.2	Differentiate between assets and liabilities.	02	2	1
	1.3	Distinguish between single entry and double entry methods of book keeping.	02	2	2
	1.4	What is the trial balance, and why is it prepared?	02	2	2
	1.5	What is the purpose of preparing a Profit and Loss Account?	02	2	3
	1.6	Name two simple adjustments made while preparing financial statements.	02	1	3
	1.7	List three elements of costing.	02	1	1
	1.8	Differentiate between financial accounting and cost accounting.	02	2	4
	1.9	What is material cost variance, and why is it important?	02	2	1
	1.10	Standard cost of material for output of 2,600 units is 71,500 rupees and actual output is 2,550 units. If material mix variance is 1,095 rupee adverse, what will be the material usage variance?	02	2	4

PART B

2	a	Explain the objectives of Financial Accounting and its significance in business decision –making.	10	2	1
	b	Briefly describe how budgeting helps in cost control and decision-making. Also, mention two types of budgets commonly used in organizations.	06	3	1
3	a	Explain the rules for journalizing the transaction with appropriate illustration.	04	3	1
	b	Journalise the following transactions in the books of trader . Debit Balance on Jan. 1, 1999: Cash in hand *Rs* 8,000, Cash at Bank *Rs* 25,000, stock of Goods *Rs* 20,000, Furniture *Rs* 2,000, and Building *Rs* 10,000. Sundry Debtors: Vijay *Rs* 2,000, Anil *Rs* 1,000, and Madhu *Rs*, 2,000. Credit Balances on Jan. 1, 1999: Sundry Creditors : Anand Rs 5,000, Loan from Bablu *Rs* 10,000 Following were further transaction in the month of January, 1999: 1. Jan. 1, Purchased goods worth *Rs* 5,000 for cash less 20 % trade discount and 5 % cash discount. 2. Jan. 4, Received *Rs* 1,980 from Viajy and allowed him *Rs* 20 as discount. 3. Jan, 6, Purchased goods from Bharath *Rs* 5,000. 4. Jan. 8, Purchased plant from Mukesh for *Rs* 5,000 and paid *Rs* 100 as cartage for bringing the plant to the factory and another *Rs* 200 as installation charges			

5. Jan. 12, Sold goods to Rahim on credit *Rs* 600.
6. Jan. 15 , Rahim became an insolvent and could pay only 50 paise in a rupee.
7. Jan. 18, Sold goods to Ram for cash *Rs* 1,000.
8. Jan. 20, Paid salary to Ratan *Rs* 2,000.
9. Jan. 21, Paid Anand Rs 4,800 in full settlement.
10. Jan. 26, Interest received from Madhu *Rs* 200.
11. Jan. 28, Paid to Bablu interest on loan *Rs* 500.
12. Jan. 31, Sold Goods for cash *Rs* 500.
13. Jan. 31, Withdrew good from business for personal use *Rs* 200.

12 | 4 | 2

OR

4 a Explain the process of posting from Journal to Ledger. 04 | 2 | 2

b Journalise the following transactions , post them in the Ledger and balance the accounts on 31st January.

- Ram started business with capital of *Rs* 10,000.
- He purchased goods from Mohan on credit *Rs* 2,000.
- He paid cash to Mohan *Rs* 1,000.
- He sold goods to Suresh *Rs* 2,000.
- He received cash from Suresh *Rs* 3,000.
- He further purchased goods from Mohan *Rs* 2,000.
- He paid cash to Mohan *Rs* 1,000.
- He further sold goods to Suresh *Rs* 2,000.
- He received cash from Suresh *Rs* 1,000.

12 | 4 | 2

5 a Prepare the Trading Account of Mr. Ramesh for the year ending 31st December , 1998 from the date as follow:

	Rs		*Rs*
Purchase	10,000	*Wages*	4,000
Purchase Returns	2,000	*Carriage charges*	2,000
Sales	20,000	*Stock on* 1.1.1998	4,000
Sales Returns	5,000	*Stock on* 31.12.1998	6,000

06 | 3 | 3

b Find out the cost of merchandise purchased , cost of merchandise sold, cost of merchandise unsold and Gross Profit from the following transactions:

	Rs.
Purchase (3,000 *articles*)	25,000
Freight	1,000
Local Taxes	1,000
Salaries	2,500
Shop Rent	500
Godown Rent	500
Electrical Charges	600
Municipal Taxes	200
Stationary	250
Furniture (*estimated life* 5 *years*)	12,000
Sales (2,700 *articles*)	32,000

10 | 4 | 3

OR

6 a Distinguish between Profit and Loss Account and Balance sheet 04 | 3 | 3

b From the following balance extracted from the book of M/s Rajendra kumar Gupta & Co., pass the necessary closing entries, prepare a Trading and Profit and Loss Account and a Balance Sheet.

Particulars	*Rs*	*Particulars*	*Rs*
Opening Stock	1,250	*Plant and Machinery*	6,230
Sales	11,800	*Returns Outwards*	1,380
Depreciation	667	*Cash in hand*	895

			Particulars	*Rs*	*Particulars*	*Rs*		
			Commission (Cr.)	211	*Salaries*	750		
			Insurance	380	*Debtors*	1,905		
			Carriage Inwards	300	*Discount(Dr.)*	328		
			Furniture	670	*Bill Receivable*	2,730		
			Printing Charges	481	*Wages*	1,589		
			Carriage Outwards	200	*Returns Inwards*	1,659		
			Capital	9,228	*Bank Overdraft*	4,000		
			Creditors	1,780	*Purchases*	8,679		
			Bill payable	541	*Petty Cash in Hand*	47		
					Bad debts	180	12	4	3
7	a	Explain the elements of costing in detail, and discuss their significance in cost accounting.					06	2	1
	b	From the following information prepare a cost sheet Company Desire a margin of 20 % on the cost of sales.							
			Particulars	*Amount in Rs*					
			Direct material – purchased	80,000					
			Direct material – Opening stock	20,000					
			Direct material – Closing stock	25,000					
			Productive wages	22,000					
			Direct expenses	5,000					
			Consumable stores	4,000					
			Factory manager salary	15,000					
			Unproductive wages	7,000					
			Factory overheads	12,000					
			Work – in – progress						
			Opening Stock	13,000					
			Closing Stock	7,000					
			Office and administrative overheads	28,000					
			Opening stock of finished goods	5,000					
			Closing stock of finished goods	10,000					
			Selling and distribution overheads	33,000			10	4	4
					OR				
8	a	Explain how cost accounting helps in managerial decision-making.					04	2	1
	b	The following particulars have been obtained from the cost records for the year. Assume that all products manufactured during the year have been sold to earn a profit of 20% on selling price.							
				Rs.					
			Material used in manufacturing	1,10,000					
			Material used in Primary packing	20,000					
			Material used in selling the product	3,000					
			Material used in factory	1,500					
			Productive Wages	30,000					
			Factory Supervision expenses	4,000					
			Material used for office	2,500					
			Chargeable Expenses	10,000					
			Indirect expenses – factory	2,000					
			Administrative expenses	3,000					
			Depreciation on Factory Building	3,500					
			Depreciation on Office Building	1,500					
			Freight on materials purchased	5,000					
			Depreciation on Delivery Van	1,000					
			Salary paid to the driver of Delivery Van	3,600					
			Advertisement	2,000					
			Bad Debt	1,500			12	4	4

9	a	Explain the components of standard cost and their role in measuring organizational performance.	08	3	4
	b	JVG Ltd. Produces a product and operates a standard costing system and value material and finished goods inventories at standard cost. The information related with the product is as follows: $Particulars$ | $Cost\ per\ unit(Rs)$ $Direct\ Materials$ $(30kg\ at\ Rs.350\ per\ kg)$ | 10,500 $Direct\ labour$ $(5\ hours\ at\ Rs.80\ per\ hour)$ | 400 The actual information for the month just ended is as follows: i) The budgeted and actual production for the month of September 2019 is 1,000 units ii) Direct materials – 5,000 kg at the beginning of the month. The closing balance of direct materials for the month was 10,000 kg. Purchases during the month were made at $Rs.365$ per kg. The actual utilization of direct materials was 7,200 kg more than the budgeted quantity. iii) Direct labour- 5,300 hours were utilized at a cost of Rs .4,34,600. Calculate : x) Direct material price and usage variances. y) Direct labor rate and efficiency variances.	08	3	4
10	a	Discuss the significance of variance analysis in standard costing. How do material cost variance , labor cost variance , and overhead cost variance contribute to effective cost control?	08	3	4
	b	Gama Ltd. Has furnished the following Standard Cost data per unit of production: • Material $10Kg@\ Rs.10$ per Kg • Labor 6 hours @ $Rs.5.50$ per hour • Variable overhead 6 hours @ $Rs.10$ per hour • Fixed overhead Rs 4,50,000 per month (Based on a normal volume of 30,000 labour hours) The actual cost data for the month: Material used 50,000 kilogram at the cost of Rs 5,25,000. • Labour Paid $Rs.1{,}55{,}000$ for 31,000 hours worked. • Variable Overheads $Rs, 2{,}93{,}000$ • Fixed Overheads $Rs.4{,}70{,}000$ • Actual Production 4,800 units Calculate: i) Material Cost Variance ii) Labour cost variable iii) Fixed overhead cost variance iv) Variable Overhead Cost variance	08	3	4

Quality Assurance

IM353IA | V Semester SEE Question Papers

- Feb/Mar 2026 (IM353IA)
- Feb/Mar 2025 (IM353IA)
- Mar/Apr 2023, Statistical Process Control (18IM53)
- Jan/Feb 2021, Statistical Process Control (18IM53)
- Jan/Feb 2021, Quality and Reliability Engineering (16IM53)
- Fast Track Oct 2020, Quality and Reliability Engineering (16IM53)
- Nov/Dec 2019, Quality and Reliability Engineering (16IM53)
- Fast Track Jul 2019, Quality and Reliability Engineering (16IM53)

Question Paper ID: 790679

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM353IA

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026
INDUSTRICAL ENGINEERING AND MANAGEMENT

Quality Assurance

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical table is permitted.
4.Use of statistical process control chart formula table is permitted.

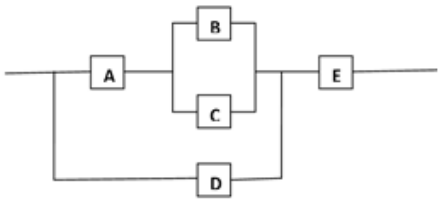
Part A

Question No	Question	M	CO	BT
1.1	The concept of integrating digital technologies with QA practices is called _____.	01	2	2
1.2	What is Quality 4.0? Mention two technologies used.	02	1	2
1.3	A control chart contains a centre line, _____ and _____.	01	1	1
1.4	Define Cpk. How is it different from Cp?	02	2	2
1.5	Define Cp. How is it interpreted	02	2	2
1.6	What is AOQL?	02	2	2
1.7	What is AOQ?	02	2	2
1.8	What is replication and why is it important?	02	2	2
1.9	What is a factorial experiment?	02	3	2
1.10	What is failure rate?	02	2	2
1.11	Differentiate between repairable and non-repairable systems.	02	2	2

Part B

<table>
<tr><th>Question No</th><th>Question</th><th>M</th><th>CO</th><th>BT</th></tr>
<tr><td>2a</td><td>Discuss any four quality dimensions with examples.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>2b</td><td>Explain any four statistical concepts used in control charts.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>3</td><td>In a foundry the specification on mould hardness is 65 ±15. Twenty-five moulds were chosen at random and mould hardness was estimated at 4 different places/mould. The data obtained are given in the following table:
<table>
<tr><th>Mould No.</th><th>$\bar{X}$</th><th>R</th><th>Mould No.</th><th>$\bar{X}$</th><th>R</th></tr>
<tr><td>1</td><td>68</td><td>5</td><td>13</td><td>70</td><td>10</td></tr>
<tr><td>2</td><td>59</td><td>16</td><td>14</td><td>67</td><td>10</td></tr>
<tr><td>3</td><td>60</td><td>15</td><td>15</td><td>63</td><td>15</td></tr>
<tr><td>4</td><td>57</td><td>12</td><td>16</td><td>69</td><td>10</td></tr>
<tr><td>5</td><td>74</td><td>24</td><td>17</td><td>62</td><td>5</td></tr>
<tr><td>6</td><td>68</td><td>5</td><td>18</td><td>56</td><td>10</td></tr>
<tr><td>7</td><td>66</td><td>10</td><td>19</td><td>66</td><td>10</td></tr>
<tr><td>8</td><td>56</td><td>10</td><td>20</td><td>63</td><td>8</td></tr>
<tr><td>9</td><td>78</td><td>15</td><td>21</td><td>67</td><td>15</td></tr>
<tr><td>10</td><td>55</td><td>10</td><td>22</td><td>75</td><td>10</td></tr>
<tr><td>11</td><td>67</td><td>5</td><td>23</td><td>53</td><td>5</td></tr>
<tr><td>12</td><td>57</td><td>5</td><td>24</td><td>56</td><td>12</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
i). Construct the control chart for the data
ii).Interpret the state of the manufacturing process
iii).Can the process meet the specification?
iv).What is the current state of affairs regarding non-conformity?
v). What are the different possible ways of reducing the rejection?</td><td>16</td><td>4</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>Explain the steps in Construct $\bar{X}$–R charts. and interpret $\bar{X}$–R charts.</td><td>10</td><td>3</td><td>2</td></tr>
<tr><td>4b</td><td>What are Process capability index Cp and Cpk? Explain their significance.</td><td>06</td><td>2</td><td>3</td></tr>
<tr><td>5a</td><td>Differenciate between Producer’s Risk vs Consumer’s Risk,AQL and LQL and their Importance</td><td>10</td><td>3</td><td>3</td></tr>
</table>

5b	Explain the following indices used in acceptance sampling plans i) Acceptable Quality Level ii) Rejectable Quality Level	06	3	3
	OR			
6a	What are the characteristics of a good acceptance sampling plan	06	3	2
6b	A double sampling plan is as follows: Inspect a first sample of 25 items randomly form a lot. The lot is accepted if no defectives are found. If there or more found, the lot is rejected. Otherwise a second sample of 100 is drawn and the lot is accepted if the combined number of defectives found does not exceed. i) Determine the producer's risk (a) for a lot fraction defective of 0.01 ii) Determine the consumer's risk (b) for a lot fraction defective of 0.07 iii) What is the ATI of the above plan for a 2% defective lot.	10	2	3
7a	Discuss with examples basic principles of experimental design.	08	2	2
7b	Explicate with illustrations the strategy of experimentation.	08	2	2
	OR			
8a	Define Randomized Block Design RBD. give Example	04	2	1
8b	How do the principles of replication, randomization, and local control contribute to the validity and reliability of experimental results in design of experiments?	10	2	2
8c	What is interaction in factorial design?	02	2	2
9a	A participant in a motor rally is required to complete a mission of 2500 km. His vehicle can be imagined to have 3 subsystems, namely, fuel injection system, ignition system and other systems. The mean distance to repair the 3 subsystems is known to be 6000 kms, 8000 kms & 10000 kms respectively. Find the reliability of motorist completing mission without repair.	08	3	4
9b	The MTBF of a certain unit is 100 hours. Calculate the reliability of the unit for 150 hoursof operation. If the reliability of the units is to be increased by 15%, 25%, 35%,	08	3	4

	45% and55%. Calculate the corresponding % changes in the MTBF that is necessary. Plot % changes in reliability vs % MTBF			
	OR			
10a	Distinguish between: i Repairable and Irrepairable systems; ii Parallel and Series systems.	08	3	3
10b	Consider the following series-parallel configuration:  Determine the reliability of the system for 20 hours of operating period. The configuration is given below. The failure rate / hour: $\lambda_A = 0.01$, $\lambda_B = 0.0125$, $\lambda_C = 0.02$, $\lambda_D = 0.02$, $\lambda_E = 0.025$.	08	3	3

IM353IA

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

QUALITY ASSURANCE

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	Differentiate between Quality Control (QC) and Quality Assurance (QA)	02	2	1
	1.2	What is the chance and assignable causes of variation?	02	2	1
	1.3	What is the primary objective of ISO 14000 standards?	02	1	1
	1.4	What are the main objectives of a quality audit?	02	2	1
	1.5	List any four dimensions of quality.	02	1	1
	1.6	Expand AQL and $AOQL$.	02	1	1
	1.7	What is a completely randomized design (CRD) in experimentation?	02	2	1
	1.8	What is the primary advantage of using factorial experiments?	02	2	1
	1.9	A system consists of two components arranged in parallel. The reliabilities of the components are $R1 = 0.80$ and $R2 = 0.85$. What is the overall reliability of the system?	02	3	1
	1.10	What does $MTTF$ stand for and why is it important?	02	1	1

PART-B

2	a	Explain the concept of Rational Subgrouping and its principles. Apply these principles to real-world scenarios in manufacturing or service industries.	08	2	1
	b	Classify the Cost of Good Quality and Cost of Poor Quality in the automotive industry. Provide examples of each cost category and explain how they can be managed effectively to improve the product performance and reduce overall expenses.	08	2	1
3	a	What are the two most common types of control charts for variables? Briefly explain their purpose.	06	2	1
	b	Lots of Bushing screws from a single supplier were inspected 100 percent with the following results.			

Lot No	*Amount in lot*	*Number rejected*
1.	6000	183
2.	6000	131
3.	10000	184
4.	1783	23
5.	3089	171
6.	5774	65
7.	3000	46
8.	2724	85

Draw an appropriate control chart for the data. What can you conclude on the state of control? 10 3 3

OR

4 a What are the two most common types of control charts for attributes? Briefly explain their purpose. 06 2 1

b A manufacturing company measures the diameter of a component in millimeter during a production process. The data are collected in 5 samples, each with 4 measurements. The data are as follows:

Sample	*Measurement* 1	*Measurement* 2	*Measurement* 3	*Measurement* 4
1.	10.2	10.3	10.1	10.4
2.	10.6	10.8	10.4	10.5
3.	10.4	10.2	10.2	10.3
4.	10.1	10.3	10.3	10.2
5.	10.5	10.4	10.6	10.5

i) Calculate the Mean ($\bar{X}$) and Range (R) for each sample.
ii) Determine Control Limits using standard factors for a sample $A2 = 0.729, D3 = 0, D4 = 2.282$.
iii) Interpret the Control Chart. 10 3 3

5 a With the help of a OC curve, define the terms producer's risk and consumer's risk in acceptance sampling plan. 06 2 1

b Draw the OC curve, AOQ curve, ASN curve and ATI curve for the single sampling plan with $N = 5000, n = 100$ and $c = 2$. 10 3 2

OR

6 a Compare single, double and multiple sampling plans. 06 2 1

b Suppose a consumer receives lots of 500 candles from a new supplier. To check the quality of the lot, the consumer draws one sample of size 20 and accepts the lot if the inspected sample contains at most one defective candle. Otherwise he/she rejects the lot. Construct the OC curve for this plan. 10 3 2

7 a Explain the purpose and procedure of a "Single Factor Experiment" in the context of DOE. How is this type of experiment different from a "Multi-Factor Experiment" and when would each be used? 04 2 1

b A chemical engineer has conducted an experiment to study the effect of temperature and pressure on the reaction time of a chemical process. Let the two factors temperature and pressure are denoted by A and B respectively. Suppose the following results are obtained from this study. The number of replications is two.

$Pressure(B)$	$Temperature\ (A)$	
	$Low(60\,℃)$	$High(70\,℃)$
$Low(100MPa)$	40	43
	37	50
$High(150\,MPa)$	59	37
	54	43

Estimate the factor effects. Which effects are large?
Perform an analysis of variance. What are your conclusions? 12 3 4

OR

8 a Explain the concept of "Randomization" in the context of the design of experiments. Why is randomization important and how does it help in eliminating bias and improving the validity of the experimental results? Provide an example of how randomization can be applied in a real-worlds experiment. 08 2 4

	b	A researcher is testing the effect of three different fertilizers (AAA, BBB and CCC) on crop yield (in kg). The researcher conducts an experiment using a randomized block design with 4 replicates per fertilizer type. The observed yields are as follows: $\begin{array}{\|c\|c\|}\hline \boldsymbol{Fertilizer} & \boldsymbol{Yield(kg)} \\ \hline AAA & 22, 24, 20, 23 \\ \hline BBB & 28, 27, 29, 30 \\ \hline CCC & 25,26,24,27 \\ \hline \end{array}$ Perform a one-way $ANOVA$ to determine if there is a significant difference in mean yield among the fertilizers at a 5% significance level.	08	3	3
9	a	Stare the General failure rate curve (Bath tub curve) of an electronic product and identify the causes of failures in the various zones.	08	2	2
	b	In a rerolling process 2 blowers are connected to reheat the furnace using pulverized coal. One of the blowers is standby equipment. The blower system includes: Stator, motor, coupling and blower with the following reliabilities: $0.9, 0.92, 0.85$ and 0.95 respectively. The pulverized coal system has a reliability of 0.99 and the furnace of 0.95 for an operating period of 250 hours. Draw the system configuration and determine the reliability of the total system for 250 hours of operation.	08	3	4
		OR			
10	a	Define the following terms: Reliability, $MTTF, MTBF, MTTR$, Availability and Maintainability.	10	2	1
	b	An amplifier has an exponential time to failure distribution with a failure rate of 10% per 2000 hours. What is the reliability of the amplifier at 8000 hours? Find the mean time to failure.	06	3	3

18IM53

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations March /April -2023

Industrial Engineering and Management

STATISTICAL PROCESS CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Tables permitted.

PART-A

1	1.1	Give any one of the definitions of quality.	01
	1.2	What is the broad purview of ISO 9000 standards?	01
	1.3	Consider an operation on a CNC machine. Give an example of a possible assignable cause.	01
	1.4	What is the proportion of the Normal distribution that is covered under the 35 control limits?	01
	1.5	Define the term warning limit.	01
	1.6	What can you conclude when a point falls below the lower control limit on a control chart for fraction non-confirming chart (p-chart)?	01
	1.7	Consider the process of costing. Give an example of a defect.	01
	1.8	Write down the formula for process performance index C_{PK} taxing into account process centering.	01
	1.9	Mention any one of the uses of the chart for individual measurements.	01
	1.10	When do you prefer to use a group control chart for monitoring the process quality?	01
	1.11	Define the term producer's risk.	01
	1.12	Give an example of any one of the published sampling plans.	01
	1.13	Graphically indicate the presence of an interaction effect between two factors.	01
	1.14	What is the meaning of the term randomization?	01
	1.15	How many experimental runs are to be carried out in the case of a 2^5 factorial design?	01
	1.16	What is the meaning of a full factorial design?	01
	1.17	Sketch the time versus failure rate curve for the components failure according to exponential distribution.	01
	1.18	Give a practical example of a series device configuration.	01
	1.19	If there are 'n' components in parallel configuration, all the components are identical. What is the formula for estimating the reliability of the system, given that the reliability of each component in R?	01
	1.20	Why the common failure rate curve is also called as the bath tub curve?	01

PART-B

2	a	Take a simple product and identify the 8 dimensions of quality. Discuss how these dimensions determine the quality of this product.	08
	b	What is the true meaning of a six sigma process? List out the percent inside specifications and the defectives in terms of PPM level for sigma limits of specifications from 1 to 6. Assuming normal distribution and a perfectly centered process. What happens to these two performance measures when there is a shift in the mean of the process to the extent of ± 1.55 from the target? Discuss.	08
3	a	Sample of $n = 6$ items each are taken from a process at regular intervals. A quality characteristic is measured and $\bar{X}$ and R are calculated for each sample. After 50 samples, we have $\sum_{i=1}^{50} \bar{X}_i = 2000$ and $\sum_{i=1}^{50} R_i = 200$. Assume that the quality characteristic is normally distributed. i) Compute the control limits for the $\bar{X}$ and R charts. ii) All points on the control charts fall between the control limits computed in part (i). What are the natural tolerance limits to the process? iii) If the specification limits are 41 ± 5.0, what are your conclusions regarding the ability to the process to produce items within these specifications? iv) Assuming that if an item exceeds the upper specification limit it can be reworked and if it is below the lower specification limit, it must be separated. What percent of scrap and rework is the process producing?	12
	b	Briefly outline any one of the methods of estimating process capability.	04
		OR	
4	a	A process is being controlled with a fraction non-confirming control chart. The process average has been shown to be 0.07. Three-sigma control limits are used and the procedure calls for taking daily samples of 400 items. i) Calculate the upper and lower control limits. ii) If the process average should suddenly shift to 0.10, what is the probability that the shift would be detected on the first subsequent sample?	08
	b	A control chart for non-conformities is maintained on a process producing desk calculators. The inspection unit is defined as two calculators. The average number of non-conformities per machine when the process is in control is estimated to be two. i) Find the appropriate three sigma control limits for this size inspection units. ii) What is the probability of Type-I error for this control chart?	08
5	a	Briefly outline the basic principles of the cusum control chart for monitoring the process mean.	04
	b	Suppose that a supplier ships components in lots of size 5000. A single sampling plan with $n = 50$ and $C = 2$ is being used for receiving inspection. Rejected lots are screened and all defective items are reworked and returned to the lot.	

		i) Draw the OC curve for this plan. ii) Find the level of lot quality that will be rejected 90% of the time. iii) Management has objected to the use of the above sampling procedure and wants to use a plan with an acceptance number $C = 0$ arguing that this is more consistent with their zero defects program. What do you think of this?	12
		OR	
6	a	Give an example of any three situations in which the sample size used for process monitoring is n=1; that is the sample consists of an individual unit is used.	04
	b	Draw the primary and supplementary OC curves for a double sampling plan with $n_1 = 50, C_1 = 2, n_2 = 100, C_2 = 6$. If the incoming lots have fraction non-conforming$p = 0.05$, what is the probability of acceptance on the first sample? What is the probability of final acceptance? Calculate the probability of rejection on the first sample.	12
7	a	An experiment was run to determine whether four specific fusing temperatures affect the density of a certain type of brick. The experiment led to the following data.	

Temperature	Density				
100	21·8	21·9	21·7	21·6	21·7
125	21·7	21·4	21·5	21·4	
150	21·9	21·8	21·8	21·6	21·5
175	21·9	21·7	21·8	21·4	

		i) Does the fusing temperature affect the density of the bricks? Use $\alpha = 0.05$ ii) Analyze the residuals from this experiment.	12
	b	Briefly outline the guidelines for designing experiments.	04
8	a	Define the following terms. i) Reliability ii) Failure rate iii) $MTBF$	03
	b	A component has a failure rate which is according to experimental failure law which is $\lambda = 4 \times 10^{-3}\ failures/hour$. i) Determine the reliability of this component for $t = 10hrs, 100hrs, 1000hrs, 10000hrs$. ii) Plot the time versus reliability graph. Comment on the plot that you have obtained.	05
	c	Three components are in series with failure rate as follows. Component 1 failure rate $\lambda_1 = 4 \times 10^{-6}\ failures/hour$ Component 2 failure rate $\lambda_2 = 4 \times 10^{-4}\ failures/hour$ Component 3 failure rate $\lambda_3 = 6 \times 10^{-6}\ failures/hour$ Determine the reliability of this series system configuration. If the three components were to be arranged in parallel configuration, what would be the system reliability? (Assume $t = 1000hr$).	08

18IM53

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb – 2021

Industrial Engineering and Management

STATISTICAL PROCESS CONTROL

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Tables is permitted.

PART-A

1	1.1	What is the first step in Quality Assurance?	01
	1.2	The statistical process control helps identify _____ type of causes that affect the process.	01
	1.3	In Six Sigma, the value of $C_{p=}$ _____.	01
	1.4	_____ is a method to find factor effects and interaction effects.	01
	1.5	Effect of one factor depends on where some other factor is set. This is called as _____.	01
	1.6	The mean of sampling distribution is equal to _____ of process distribution.	01
	1.7	The percent of the sample means will have values that are within ±3 standard deviations of the distribution mean is _____.	01
	1.8	In control charts, the dividing lines between random and nonrandom deviations from mean of the distribution are known as _____.	01
	1.9	Central tendency of a process is monitored in _____ control chart.	01
	1.10	The control chart used for the number of defects per unit is _____.	01
	1.11	The costs related to formal technical reviews and test equipment reviews are related to _____ type of quality cost.	01
	1.12	In double sampling plan, if the number of defects is in between the two cut off numbers C_1 *and* C_2 then the next step is to _____.	01
	1.13	In _____ sampling plan, the samples are accumulated at each stage till the decision of accepting or rejecting a lot is taken.	01
	1.14	In addition to OC curve, _____ and _____ curves are used to select an appropriate sampling pan.	02
	1.15	To find out whether noise affects the capability to solve the assignment of Quality Management, one group solve the assignment in a silent room and another group solve the assignment in a noisy room. The group solving problems in the silent room completes 22 problems in one hour and the group solving problems in the noisy room completed 15 problems in one hour. In this experiment, the independent variable is _____ and the dependent variable is _____.	02

	1.16	What statistical procedure is used to assess the statistical significance of the main effects and the interaction(s) in a factorial design? What is the test statistic used for this?	02
	1.17	One of the concepts of reliability is _____ which is defined as the probability that the system is operating properly when it is requested for use.	01

PART-B

2	a	How is the application of concepts of statistical quality control advantageous over 100% inspection? Explain.	10
	b	Explain the need for the change in the control limits of the control charts.	06
3	a	Identify and explain the cause of i) Trend ii) Cyclic pattern in control charts.	06
	b	Mean values and ranges of data from 20 samples (sample size =4) are shown in the table below: Calculate the control limits for x-bar and R charts. Draw the control charts for the same and give suitable conclusions.	10

S.N.	Mean	Range	S.N.	Mean	Range
1	10	4	11	12	5
2	15	4	12	13	4
3	12	5	13	12	4
4	11	4	14	12	3
5	9	5	15	11	3
6	11	6	16	15	4
7	11	4	17	12	4
8	9	4	18	15	3
9	10	4	19	11	3
10	11	6	20	10	4

OR

4	a	Ten samples of 15 components are tested with results below. Construct p-chart and draw conclusions.	08
	b	A stable normally distributed process has the specification 3.5 ± 0.03, given $\mu = 3.51$ and standard deviation = 0.016, determine $C_p and\ C_{pk}$. What conclusions can you draw about the capability of the process?	04
	c	Discuss any four principles of rational sub-grouping.	04

Sample	No. of defectives
1	2
2	1
3	3
4	2
5	3
6	1
7	0
8	1
9	0
10	1

5 Consider the following data of a double sampling plan. Construct Operating Characteristic Curve, AOQ curve. Also determine ATI and ASN.

N = 5000

n1 = 150 n2 = 200

c1 = 3 c2 = 8

r1 = 6 r2 = 9 — 16

OR

6 a What are the assumptions that you make while designing a single sampling plan? — 05

b Compare and contrast single, double and triple sampling plans. — 05

c How Type A and Type B OC Curves are different? When are they used? — 06

7 a Differentiate between CRD and RBD. — 04

b The data below is from a cattle experiment in which temperatures, protein and gender were evaluated as to efficiency of weight production (feed per pound of gain). Two replicates of each treatment were run according to a Completely Randomized Design. Conduct ANOVA for main effects and interaction effects. Draw suitable conclusions. — 12

House Temp	90°				60°			
Protein Level	High		Low		High		Low	
Gender	M	F	M	F	M	F	M	F
Replication 1	2.96	2.69	2.8	2.6	2.89	2.73	2.94	2.85
Replication 2	3.04	2.75	2.72	2.7	2.75	2.67	3	2.75

8 a Compare and contrast series and parallel configurations of system reliability. Give suitable examples. — 04

b A system has three units with a failure rate of 1.5, 4, 3.8 failures per 10^6 hours respectively. Determine.

i) MTBF of the system.

ii) The reliability of the system for 10 hours if the components are connected in series.

iii) Reliability if these components are connected in parallel. — 08

c What do you think are the causes that lead to high failure during infant mortality period? — 04

16IM53

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb-21

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Quality cannot be inspected into product nor can it be advertise in, it must be designed and built in. Comment on the statement.	02
	1.2	A physical or chemical property or any other requirements used to define the nature of the product is termed as _____.	01
	1.3	State Central limit theorem.	01
	1.4	Differentiate between chance and assignable causes of variation.	02
	1.5	The usual purpose of an R-chart is to signal whether there has been a gain or loss in _____.	01
	1.6	_____ plans details the action to be taken once an out of control situation is detected.	01
	1.7	Differentiate between defect and defective.	02
	1.8	The local newspaper receives several complaints per day about typographic errors. Over a seven-day period, the publisher has received calls from readers reporting the following number of errors: 4,3,2,6,7,3 and 9. Based on these data alone,. What type of control chart(s) should the publisher use ?	02
	1.9	The number of invoices that contain errors is an example of _____ data.	01
	1.10	Average total inspection under single sampling plan in which rejected lots are 100% inspected is given by the formula ATI= _____.	01
	1.11	State the three parameters of the single sampling plan.	02
	1.12	The uncontrollable input factors that affect the output process are also termed as _____.	01
	1.13	The experiment in which two factors are varied at two levels is called as _____ experiment.	01
	1.14	The probability that the device/equipment will be restored to its operational effectiveness with the given period is called as _____.	01
	1.15	What is the difference between an experimental and a control group?	01

PART-B

2 a Identify the various dimensions of quality dimensions as proposed by Garvin. Explain their significance with respect to product quality taking suitable examples. 08

b What do you mean by Quality cost? Why is it so important? Differentiate between Failure cost and appraisal cost. 08

3 a In a manufacturing unit of certain transformers the units are required to meet a number of specifications and are subjected to inspection daily. At the end of twenty days 230 units have been rejected out of 4150 produced and inspected. Determine

i. The 3 sigma control limits based on any average daily production of 200 units

ii. Only one point has crossed the control limits. On that day 30 non-conforming items were found in 200 units inspected.

iii. What is the aimed value of p?

iv. What control limits would you recommend for future use? 10

b Distinguish clearly between

(i) P and U charts

(ii) C and U charts. 06

OR

4 a The following $\overline{X}$ – R values of 20 subgroups of 5 readings.

Sub group	$\overline{X}$	R	Subgroup	$\overline{X}$	R
1	34.0	4	11	38.4	4
2	31.6	2	12	34.0	14
3	30.8	3	13	35.0	4
4	33.8	5	14	33.8	7
5	31.6	3	15	31.6	5
6	33.0	5	16	33.0	7
7	28.2	13	17	32.6	3
8	33.8	19	18	31.8	9
9	37.8	6	19	35.6	6
10	35.8	4	20	33.0	4

(i) Determine the control limits of $\overline{X}$ and R chart

(ii) What is the process capability?

(iii) Is the process capable of meeting the specification limits ?

(iv) Determine the percentage of rejection if any. (Note : The specification limits are 33 ± 5)

(v) What values of $\overline{X}$ and R do you recommend for future use? 16

5 a A single sampling plan is as follows : n=80, c=2

(i) Plot type – B OC curve (use distribution which is theoretically correct)

(ii) If AQL is 1.5% find the producer's risk

(iii) If consumer's risk is 15% find LTPD 10

b Sketch the ideal operating characteristic curve of a sampling plan. Explain its significance. 03

c Explain the difference between average sample number and average total inspection. State any assumptions made. 03

		OR	
6	a	A double sampling plan is as follows N=5000 n1=150 c1=3 r1=6 n2=200 c2=8 r2=9 Lot fraction defective is 1.5% . Calculate Pa, AoQ, ATI and ASN	12
	b	Differentiate between type A and type B OC curve.	04
7	a	Explain the utility of "bath tub curve". What are the reasons for failure during the three stages of equipment life?	08
	b	A step down transformer, rectifier, filter comprises a series of systems. The following are the failure rates of these components. Transformer =1.56 percent failures / 1000 hours Amplifier = 2 percent failure / 1000 hours Filter = 1.7 percent failures / 1000 hours The equipment is operated for 1500 hours. What is the probability of survival of the system?	08
8	a	Determine the factors, probable levels of a factor and response for the situations given below. i) The Indian cricket team wants to determine which combination of playing XI will enhance the team's performance in the World Cup matches. ii) A casting company which desires to produce Good Quality Castings.	10
	b	Define the following terms with an example: (i) Factor (ii) Level (iii) Main effect (iv) Interaction effect (v) Treatment Combination	06

16IM53

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

QUALITY AND RELIABILTY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Distinguish between chance assignable cause of variations.	02
	1.2	State any two principles of rational sub grouping.	02
	1.3	Sketch the extreme variation patterns of assignable cause.	02
	1.4	Control unit for R chart are _____ & _____.	02
	1.5	Write any four advantages of sampling technique.	02
	1.6	Write OC curve for single sampling plan.	02
	1.7	Differentiate between consumer risk and producer risk.	02
	1.8	Measure of process performance in terms of the nominal & target value is provided by C_{pk} & the mathematical expression is _____.	02
	1.9	Define level & factors.	02
	1.10	Expand $MTTF$ & $MTBF$.	02

PART-B

2	a	What are the eight dimensions of quality? Discuss with an example any two dimensions.	08
	b	List and explain major quality cost categories.	08
3	a	Sample size $n = 5$ are taken from a manufacturing process every hour. A quality characteristics is measured and $\bar{x}$ and R are computed for each sample after 25 samples have been analyzed we have $\sum_{i=1}^{25} \bar{x} = 66250$ & $\sum_{i=1}^{25} R_i = 900$. The quality characteristics is normally distributed i) Find control unit for $\bar{x}$ and R charts ii) Assume that both chart exhibit control if the specifications are 26.4 ± 0.50. Estimate the fraction non-conforming iii) If the mean of the process were 26.4 what fraction non-conforming would result.	08
	b	Sample of $n = 5$ units are taken from a process every hour. The $\bar{x}$ and R values for a particular quality characteristics are determined. After 25 samples have been collected. We calculate $\bar{x} = 20$ and $R = 4.56$ i) What are the three sigma control limits ii) Both curve exhibit control, Estimate the process standard deviation iii) Assume that the process output is normally distributed, if the specifications 19 ± 5. What are your conclusions regarding the process capability? iv) If the process mean shifts to 24. What is the probability of not detecting this on the first subsequent samples?	08

OR

4 a A process is being controlled with fraction non conforming control chart. The process average has been shown to be 0.07. Three sigma control unit are used and the procedure calls for taking daily samples of 400 items.
i) Calculate the upper & lower control unit.
ii) If the process average should suddenly shift to 0.10. What is the probability that shift would be detected on first subsequent sample
iii) What is the probability that the shift in part (ii) would be detected on the first & second sample takes after the shift? 08

b What are the differences between
i) p chart & np chart
ii) c chart & u chart 08

5 a A single sampling plan of $n = 200, c = 4$ & N is very large when compared to the sample size. Draw an AOQ curve. Find $AOQL$ after computing the probabilities of acceptance of lot of 0.5%, 1%, 2%, 3%, 4% & 5% defectives 08

b Construct an OC curve for a double sampling plan of $N = 2600$, $n_1 = 100$, $c_1 = 1, n_2 = 200$ & $c_2 = 4$. 08

OR

6 a In a single sampling plan $N = 10{,}000$, $n = 300, c = 1$ is used with the stipulation that all rejected lots are to be screened, compute the AOQ, if all lots submitted are 0.50 defective. 08

b Draw the AOQ curve for the data $N = 4000, n = 80, c = 2$. 08

7 a An article in industrial quality control describe an experiment to investigate the effect of glass type and phosphor type of the brightness of a television tube. The response measured in the current necessary (in microamps) to obtain a specific brightness level. The data are shown here. Analyze the data and draw conclusions.

Glass type	*Phosphor type*		
	1	2	3
1	280	300	290
	290	310	285
	285	295	290
2	230	260	220
	235	240	225
	240	235	230

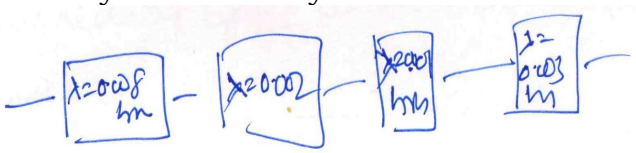
10

b Write the test matrix for the 2^2 factorial design. 06

8 a Suppose that a certain software product has a mean time between failure of 10,000 hours and has a mean time to repair of 20 hrs. If the product is used by 100 customers. What is availability? 06

b Find the reliability of the system with three components A, B & C in parallel. The reliability of A, B & C are 0.95, 0.92 & 0.90 respectively. 06

c The components in the system below are exponentially distributed with the indicated failure rates. Develop an expression for the reliability of the system. What is the system reliability at the time of 100 hrs?



06

16IM53

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of statistical table is permitted.

PART-A

1	1.1	Write different definitions for quality.	02
	1.2	Define reliability.	02
	1.3	Write any four benefits of ISO 9000.	02
	1.4	List 7 QC tools.	02
	1.5	When is type-A OC curve used?	02
	1.6	Define average outgoing quality.	02
	1.7	When do you use attribute control charts?	02
	1.8	When do you use Cusum chart?	02
	1.9	List two applications of statistical experimental design in engineering design.	02
	1.10	Consider a system consisting of three components in parallel, each with a reliability of 0.65. Find the reliability of the system.	02

PART-B

2	a	List and explain the eight dimensions of quality.	08
	b	List and explain four categories of quality cost.	08
3	a	What is variability? Distinguish clearly chance causes and assignable causes of variation.	08
	b	Explain the principle of rational subgrouping.	04
	c	Write a note on process capability study.	04
		OR	

4	a	Subgroup of 5 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$ and R values compared for each subgroup. After 25 subgroups, $\sum \bar{X} = 357.5 \qquad \sum R = 8.8$ Assume that all the points are within the control limits on both the charts. The specifications are 14.4 ± 0.4 i) Compute the control limits for $\bar{X}$ and R chart ii) What is the process capability? iii) Determine the percentage of rejections if any. iv) What can you conclude regarding its ability to meet the specifications?	12
	b	When do we use moving range chart? Explain.	04
5	a	The inspection results in a machine shop based on sampling size of 50 are given below.	

Serial Number	*Number of defectives*	*Serial Number*	*Number of defectives*
1	6	11	3
2	3	12	7
3	1	13	1
4	2	14	15
5	12	15	4
6	6	16	18
7	4	17	3
8	7	18	2
9	1	19	6
10	8	20	7

		i) Calculate the control limits for the P-chart using 3σ limits. ii) Plot the data and offer your comments on the behavior of the process. iii) What standard fraction defectives you recommend for the future period.	12
	b	Distinguish between P-chart and C-chart.	04
		OR	
6	a	A single sampling plan is as follows: $N = 5000 \qquad n = 100 \qquad C = 3$ i) Plot the OC curve. ii) Determine the AQL and $LTPD$ for $P_r = 10\,\%$ and $C_r = 15\,\%$ respectively. iii) What is the ATI of the above plan for 1.5 % defectives of the incoming lots?	10
	b	Write the flowchart for double sampling plan and explain with an example.	06

7	a	Discuss with examples the principles of experimental design.	04
	b	The quality control department of an automobile company is studying the effects of several factors on the surface finish of a spindle that is manufactured using a Lathe. Three operators and two machines were selected and two spindles were produced under each combination of the experimental conditions. The results are shown in table. Assume a significance level of 0.05 i) Write the model of $ANOVA$ of this reality ii) Check the significance of each of the components of the model $operator$ / $machine$: 1: 23, 27; 25, 26 — 2: 36, 34; 36, 39 — 3: 24, 38; 37, 34	12
8	a	Explain failure rate curve/bath tub curve with a neat sketch.	09
	b	A step down transformer, rectifier and filter constitutes a series system. The following are the failure rate of these components. Transformer =1.56 % $failure/$ 1000 $hours$ Rectifier =2 % $failure/$ 1000 $hours$ Filter = 1.7 % $failure/$ 1000 $hours$ The equipment is to operate for 1500 $hours$. What is the probability of survival of the system?	07

16IM53

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. FastTrack Examinations July-19

Industrial Engineering and Management

QUALITY AND RELIABILITY ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical and control chart tables permitted.

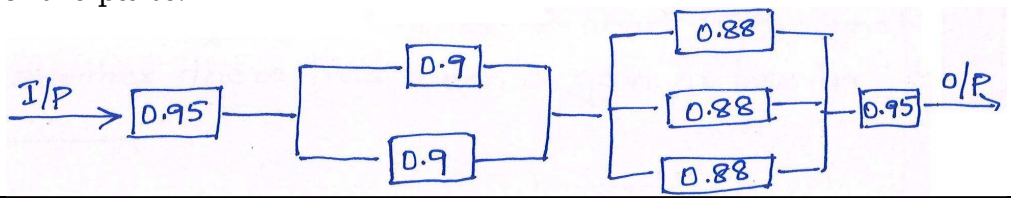
PART-A

1	1.1	The _____ are the means of translating the term 'fitness for use' into the language of technologists.	01
	1.2	The cost associated with measuring, evaluating or auditing the product comes under the costs of _____.	01
	1.3	Statethe central limit theorem	01
	1.4	Differences among materials, workers and machines is _____ type of cause of variation	01
	1.5	The usual purpose of an R-chat is to signal whether there has been a gain or loss in _____.	01
	1.6	Jars of pickles are sampled and weighed. Sample measures are plotted on control charts. The ideal weight should be precisely 11oz. which type of chart(s) would you recommend?	01
	1.7	The factors A_1, A_2, D_1, D_2, D_3 and D_4 etc. to calculate control limits depends upon _____.	01
	1.8	The local newspaper receives several complaints per day about typographic errors. Over a seven-day period, the publisher has received calls from readers reporting the following number of errors: $4, 3, 2, 6, 7, 3$ and 9. Based on these data alone, what type of control chart(s) should the publisher use?	01
	1.9	In a lot of 50 parts, 4 parts were rejected because of 12 non-conformities found in them. The proportion defectives is _____ and the average number of non-conformities per part in the lot is _____.	02
	1.10	The three parameters used for a single sampling plan are _____, _____ and _____.	02
	1.11	$MIL-STD-105E$ plans are based on _____.	01
	1.12	The uncontrollable input factors that affect the output of a process are termed as _____.	01
	1.13	In the experimental designs, the process in which every experimental unit has the same chance of being allocated to treatments s known as _____.	01

	1.14	If there are 'm' levels for a factor 'A' and 'n' levels for factor 'B' in a factorial design of experiments, then there shall be _____ combinations	01
	1.15	If 'θ' is $MTBF$, then failure rate λ is equal to _____ and reliability of the unit over the time 't' is equal to _____.	02
	1.16	Stand-by redundancy produces _____ reliability compared to parallel redundancy.	01
	1.17	Early failure of component is also called as _____.	01

PART-B

2	a	Define 'quality'. Giving two examples each, explain the five quality characteristics	06
	b	"Spending on prevention of poor quality leads to decrements in appraisal costs". Do you agree with this statement why or why not? Discuss this with the focus on components of these quality costs.	04
	c	Define and explain the type I and type II errors in the context of control charts. Are they related? How does the choice of control limits influence these two errors?	06
3	a	Subgroup of 4 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$, R values are computed for each subgroup. After 25 subgroup, $\sum \overline{X} = 15350, \sum R = 411.4$. i) Compute the control limits for $\bar{X} - R$ chart ii) Assume all the points are falling within the control limits on both the charts. The specification limits are 610 ± 15. If the quality characteristic is normally distributed what % of product would fail to meet specifications iii) Any product that falls below LSL will be scrapped and above USL must be reworked. It is suggested that the process can be centered at a level so that not more than 0.1% of the product will be scrapped. What should be the aimed value of $\bar{X}'$ to make the scrap exactly 0.1%	12
	b	A cloth manufacturing company is maintaining 100% inspection of all the finished fabric. The past 8 months data from the inspection indicated that the average number of defects/unit of fabric is 8. i) Determine the 3σ control limits ii) With these control limits, what is the probability that the process is out of control when it is really not.	04
		OR	
4	a	An item is made in large lots and a sample of 200 items are drawn and given 100% inspection. The record sheet for the first 25 samples showed that a total of 75 items were defective. i) Determine the trial control limits for np-chart ii) Assuming that all the points fall within control limits, what is the process average fraction defective? iii) If the 'p' remains constant, what is the probability that the 26th sample will contain exactly 7 defectives. iv) What is the probability that 26th sample will contain 7 or more defectives.	10

	b	What is process capability? Explain the procedure of computing process cpacnbilty using average range method?	06
5		A single sampling plan is as follows: $N = 4000;\ n = 75;\ c = 2$ a) Construct the OC curve of this plan b) If AQL is 1.5%, find producers risk and if consumers risk is 10%, find $LTPD$ c) Construct the AOQ curve and determine $AOQL$ d) Find the ATI of the above plan at 1.5% defective of the incoming lot.	16
		OR	
6		The following data refers to a 'double sampling plan': $N = 50000$, $n_1 = 150, c_1 = 0, n_2 = 100, c_2 = 1$. If 1% defective lots are submitted compute a) P_a –probability of acceptance b) AOQ- Average Outgoing Quality c) ASN d) ATI- Average Total Inspection e) AFI.	16
7	a	A single factor experiment is to be designed for estimating the tensile strength of a composite material. One of the constituents of the composite material varies in the combination at 5%, 10% and 20%. Six specimens are prepared for each combination. Explain the process of data tabulation for a completely randomized design	08
	b	Explain the application of a "linear statistical model" and $ANOVA$ for the above case	08
8	a	Sketch and explain stages of 'bath tub' curve	08
	b	Calculate the reliability of the system given below from the reliability of the parts. 	08

Enterprise Information Systems (Elective)

IM355TBB | V Semester SEE Question Papers

- Feb/Mar 2026 (IM255TBB)
- Feb/Mar 2025 (IM355TBB)
- Feb/Mar 2024 (21IM55B2)

Question Paper ID: 750439

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM255TBB

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Enterprise Information Systems

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	List two fundamental ways an Enterprise Information System contributes to improving an organization's operational efficiency.	02	1	1
1.2	Define the term 'Digital Transformation' (DT).	02	1	1
1.3	Define Robotic Process Automation (RPA).	02	2	1
1.4	Name any two leading commercial software platforms (tools) used for implementing RPA.	02	2	1
1.5	Identify a major challenge or pitfall an organization faces when relying on data from multiple, non-integrated systems (data silos) for decision-making.	02	2	2
1.6	State how data from an Enterprise Information System (EIS) supports Strategic level decision-making.	02	2	2
1.7	Define Product Life Cycle Management (PLM) in relation to product development.	02	1	1
1.8	How does the implementation of an integrated EIS/ERP system (like the one used in the Sanyo case) achieve "process quality improvement"?	02	1	2
1.9	State how the Agile practice of "continuous stakeholder feedback" benefits EIS projects regarding user satisfaction.	02	4	2
1.10	Contrast the approach to handling changing requirements between the traditional Waterfall method and the Agile method in an EIS implementation.	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Explain the concept of Digital Transformation (DT). Describe the crucial role of two key technologies, Cloud Computing and Artificial Intelligence (AI), in driving this transformation within a modern enterprise.	08	1	2
2b	Describe the hierarchical role of an Enterprise Information System (EIS) in supporting decision-making across the three organizational management levels: Strategic, Tactical, and Operational.	08	1	2

3a	A manufacturing company's procurement process is complex. Apply three different methods of BPA to the procurement process (from requisition to payment). Justify how each method specifically enhances efficiency in that stage.	08	2	3
3b	A global bank wants to automate its Customer Interaction Process. Design a BPA solution that combines Customer Interaction Automation and Decision Automation. Detail the steps involved and the types of tools necessary for this solution.	08	2	3
	OR			
4a	Explain the role of RPA frameworks such as BPMN and Agile in the lifecycle of an RPA project. Why are these structured approaches necessary even when dealing with tactical automation?	08	2	2
4b	Explain the defining difference between Basic RPA and Advanced RPA (Cognitive RPA). Discuss how Cognitive RPA, through its integration with AI/ML, enables organizations to handle complex tasks that Basic RPA cannot.	08	2	2
5a	A retail chain is analyzing sales data. They observe that sales of product 'A' dropped significantly last month. Apply the four types of data analytics sequentially to investigate this issue and formulate a management response.	08	3	3
5b	A bank wants to deploy an analytical model to predict the likelihood of default for new credit card applicants. Identify the type of data analytics involved and detail the key steps a machine learning model would take to execute this analysis.	08	3	3
	OR			
6a	Explain how the integrated data from an Enterprise Information System (EIS) supports managerial decision-making across the three levels: Operational, Tactical, and Strategic.	08	3	2
6b	Analyze the critical challenge of data silos in an organization. Explain how non-integrated data silos negatively impact the use of data for reliable, cross-functional decision-making.	08	3	2
7a	Describe the organizational conflict or hindrance that can occur when an internally optimized, non-integrated ERP system attempts to coordinate processes with external supply chain partners.	08	1	2
7b	Explain why a fully integrated EIS/ERP system is considered a foundational prerequisite before deploying advanced SCM planning applications. What core issue does the ERP solve that SCM relies upon?	08	1	2
	OR			
8a	Explain how PLM extends beyond traditional product data management by integrating data from upstream (R&D) and downstream (Service/Disposal) stages of the life cycle.	08	1	2
8b	Describe the role of a central EIS/PLM system in facilitating concurrent engineering and collaborative product development. How does this system overcome the sequential delays of traditional design processes?	08	1	2
9a	A business process is undergoing rapid change during a critical ERP customization phase. Analyze the difficulty in handling this change under a strict Waterfall methodology and demonstrate how the Agile principle of "Responding to change over following a plan" provides a superior solution.	08	4	3
9b	Prescribe how the Agile practice of using User Stories and managing a Product Backlog can effectively translate the complex, ambiguous requirements of multiple business departments into clear, actionable development tasks for a new CRM system customization.	08	4	3
	OR			
10a	Explain the importance of Cybersecurity for business-critical EIS (ERP, CRM, SCM). Detail the three foundational objectives of Information Security, known as the CIA Triad. (Sketch Required)	08	4	1
10b	Define and differentiate between four common cybersecurity threats specific to EIS: Phishing, Insider Threat, Data Breach, and Ransomware.	08	4	1

IM355TBB

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	List two key features of a Customer Relationship Management (CRM) system.	02	2	1
	1.2	Mention two types of data commonly handled by EIS.	02	2	1
	1.3	Define Business Process Automation(BPA).	02	2	1
	1.4	Mention one advantage of implementing automation in business processes.	02	1	1
	1.5	What are the three types of data analytics?	02	2	2
	1.6	Define Product Lifecycle Management (PLM).	02	2	2
	1.7	How does PLM integrate with SCM?	02	2	2
	1.8	How does a web-enabled EIS improve customer satisfaction in SCM?	02	2	2
	1.9	Name two common tools used in Robotic Process Automation (RPA).	02	1	1
	1.10	What is a key characteristic of structured data?	02	2	2

PART-B

2	a	Analyze the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answers with examples.	08	2	1
	b	How can information technology support company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answers.	08	3	1
3	a	A mid-sized manufacturing company faces delays in order processing due to repetitive manual tasks such as data entry and validation. Propose a solution using Robotic Process Automation (RPA), Discuss how implementing RPA can streamline the process and improve efficiency.	08	3	3
	b	A retail company plans to automate its invoice processing system using RPA. Explain the steps involved in the RPA implementation process for this scenarios.	08	2	3
		OR			
4	a	Design a simplified approach to integrate Business Process Automation (BPA) in the inventory management process of an e-commerce company. Include key steps and expected outcomes.	08	3	3

	b	Discuss the importance of incorporating best practices in Business Process Reengineering (BPR) initiatives. Explain how organizations can identify and implement best practices to ensure successful BPR outcomes. Provide examples of best practices commonly used in BPR projects and their impact on business performance.	08	2	3
5	a	A retail chain plans to improve its inventory management system using data warehousing. Explain how a data warehousing can help in managing inventory more effectively. Include key components of a data warehouse and their roles.	08	4	2
	b	Explain how data analytics in Enterprise Information System (EIS) can be used to improve the decision-making process for a manufacturing company aiming to reduce operational costs.	08	2	2
		OR			
6	a	Explain the concept of data mining and its role in extracting actionable insights from large datasets within Enterprise Information System (EIS). Discuss the various techniques and algorithms commonly used in data mining and provide examples of how they can be applied in different business contexts.	08	4	2
	b	Discuss the significance of Customer Relationship Management (CRM) systems in Enterprise Information System (EIS) development. Explain how CRM systems facilitate customer interactions, data analysis and relationship management across various touchpoints. Provide examples of CRM implementation strategies and their impact on enhancing customer satisfaction and loyalty.	08	3	2
7	a	Explain the role and magnitude of Enterprise Information System (EIS) in Supply Chain Management (SCM). How does EIS enhance the efficiency of supply chain operations?	08	2	3
	b	Discuss the impact of web-enabled EIS/ERP systems on CM . How do these systems improve agility and responsiveness in supply chains?	08	3	3
		OR			
8	a	Compare and contrast Enterprise Information System (EIS) and Supply Chain Management (SCM) systems. How are they integrated and what unique roles do they play?	08	3	3
	b	"Product life cycle management augments communication and teams across individual departments within a company". How are they integrated and what unique roles do they play?	08	2	3
9	a	Discuss the advantages and challenges of adopting cloud-based Enterprise Information System (EIS). How does cloud technology enhance enterprise operations?	08	4	4
	b	Provide an overview of Agile project management in Enterprise Information System (EIS). How does Agile methodology improve system implementation?	08	2	4
		OR			
10	a	"Specialized information is too diverse and even redundant". Illustrate and analyze the statement.	08	4	4
	b	Discuss the role of cyber security in safeguarding Enterprise Information System (EIS) against potential threats.	08	2	4

21IM55B2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Examinations Apr/May-2024

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	_____ Characteristics is typically associated with an Enterprise Information System (*EIS*).	01	1	1
	1.2	_____ is the main benefit of EIS from a technical standpoint.	01	2	2
	1.3	What is the primary purpose of Material Requirements Planning (*MRP*)?	01	1	2
	1.4	List a common challenge faced during *ERP* implementation related to software customization.	02	2	1
	1.5	What is one of the risks associated with Business Process Reengineering?	01	2	2
	1.6	_____ level of process quality design focuses on optimizing individual tasks within a process.	01	1	1
	1.7	Write the full form of *BPM*.	01	2	1
	1.8	How does reengineering differ from incremental improvements?	01	1	2
	1.9	Enumerate the components of Business Intelligence System architecture.	02	2	1
	1.10	Which component of *EIS* provides a visual representation of key performance indicators and metrics related to the supply chain?	01	1	2
	1.11	_____ technology is often used to enhance the visibility and traceability of products in the supply chain.	01	1	1
	1.12	How does the integration of Enterprise Information Systems with the Supply Chain Network contribute to sustainability?	02	2	2
	1.13	Which technology is commonly used to track the movement of goods in real-time within the Supply Chain Network?	01	2	1
	1.14	What is a key benefit of the trend towards a reduction in implementation time in *EIS* projects?	01	1	2
	1.15	How does the adoption of open-source technologies impact Enterprise Information System (*EIS*) implementation?	01	1	2
	1.16	What does *MRP*-3 (Money Resource Planning) primarily focus on?	01	2	1
	1.17	What is a key feature of the progression aspect in next-generation *EIS* software?	01	2	1

PART-B

2	a	Sketch the demand-supply chain service provider model. Explain the concepts of Enterprise Information Systems (*EIS*).	08	2	2
	b	Provide a summary of the features of various Enterprise Resource Planning (*ERP*) modules.	08	2	1

3	a	Develop the typical flow of inputs converted to outputs through a process. Chose a process and list the alternative accomplishing the same. Design the best process.	08	3	4
	b	Enumerate the typical infrastructural processes in an organization. Cite examples.	08	2	2
		OR			
4	a	Identify the common myths about reengineering. Elucidate the conditions under which reengineering works best.	08	3	3
	b	Provide a comparative analysis of clean state and technology enabled reengineering.	08	3	3
5	a	Develop and sketch the roadmap to embrace business intelligence. Summarize the four stages in business intelligence development.	08	4	4
	b	Select three business intelligence software solution vendors and evaluate the performance features.	08	4	4
		OR			
6	a	Write a detailed note on data warehousing. Illustrate the data warehouse linkages with information and business intelligence.	08	2	2
	b	Analyze the importance of online analytical processing ($OLAP$) in business intelligence systems.	08	3	4
7	a	Critically evaluate the statement "EIS applications help in effectively delivering Supply Chain Management".	08	3	3
	b	Explain the role of web enabled Enterprise Information Systems on Supply Chain Management.	08	2	2
		OR			
8	a	Compare Enterprise Information Systems (EIS) and Supply Chain Management (SCM).	08	3	3
	b	"Product life cycle management (PLM) facilitates Enterprise Information Systems and Supply Chain Management systems integration". Illustrate and critically analyze the statement.	08	4	5
9	a	Summarize the paradigm shift across generations of Enterprise Information System software.	08	3	3
	b	Discuss how new technologies enable Enterprise Information System implementation.	08	2	1
		OR			
10	a	Elaborate the following futuristic trends in Enterprise Information System. I) Expenditure trends II) Reduction in implementation time.	08	2	5
	b	Discuss the systems development of tailored processes for newer Enterprise Information System software.	08	2	5

Advanced Decision Modelling (Elective)

IM355TBD | V Semester SEE Question Papers

- Feb/Mar 2026 (IM255TBD)
- Feb/Mar 2025 (IM355TBD)
- Mar/Apr 2023, Decision Sciences II: Stochastic Models (18IM52)
- Jan/Feb 2021, Decision Sciences II: Stochastic Models (18IM52)

Question Paper ID: 923176

Course Code: IM255TBD

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

USN | | | | | | | | | | |

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Advanced Decision Modelling

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	State the behaviors of customer in a queuing model?	02	1	2
1.2	When arrival is 3 per hour and service is 5 per hour in an M/M/1 queue, the probability of 3 persons being in the system is _____	02	1	2
1.3	What is an absorbing state?	02	1	2
1.4	Classify the states of Markov Chain.	02	1	2
1.5	What is the objective of inventory problem analysis?	02	2	2
1.6	What is the main meaning/purpose of inventory control?	02	2	2
1.7	What is the main purpose of replacement analysis?	02	2	2
1.8	State the applications of Applications of Replacement and Maintenance Models	02	2	2
1.9	What are the standard assumptions in classical sequencing problems?	02	4	2
1.10	What is Johnson's Rule?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Consider a box office ticket window being manned by a single server. Customer arrives to purchase ticket according to Poisson input process with a mean rate of 30/hr. the time required to serve a customer has an ED with a mean of 90 seconds determine: (i) Mean queue length. (ii) Mean waiting time in the system.	08	1	3

	(iii) The probability of the customer waiting in the queue for more than 10min. (iv) The fraction of the time for which the server is busy.			
2b	An M/M/1 queuing system has an average arrival rate of $\boldsymbol{\lambda}$ = 8 customers per hour and an average service rate of μ = 10 customers per hour. i) Check whether the system is stable. ii) Find the average number of customers in the system. iii)Find the average number of customers in the queue. iv)Find the average waiting time in the system.	08	1	3
3a	Explain the classification of states in Markov chains and discuss their practical significance.	08	1	2
3b	Discuss the topological properties of the graphs of the following Markov chains: (a) $P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ (b) $P = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 0 \end{bmatrix}$ (c) $P = \begin{bmatrix} 1/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ 0 & 1/5 & 4/5 \end{bmatrix}$ (d) $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (e) $P = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$	08	2	2
	OR			
4a	A system has three possible states, 0, 1 and 2. Every hour it makes a transition to a different state, which is determined by a coin flip. For example, from state 0, it makes a transition to state 1 or state 2 with probabilities 0.5 and 0.5. i) Find the transition probability matrix. ii) Find the three-step transition probability matrix. iii) Find the steady-state distribution of the Markov chain.	08	2	3
4b	A retail store analyses the loyalty behaviour of its loyalty-card customers to understand long-term retention patterns. Based on historical data, customers are classified into the following states: • State 1: Regular buyer • State 2: Occasional buyer • State 3: Lost customer Customer movement from one period to the next follows a Markov process.	08	2	3

Initially, without any discount scheme, the one-period transition probability matrix was:

$$P_0 = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0 & 0 & 1 \end{bmatrix}$$

To improve customer retention, the store introduces a new discount policy. As a result, the transition probabilities from the Regular Buyer state (State 1) change as follows:

- Probability of remaining a Regular buyer = 0.75
- Probability of becoming an Occasional buyer = 0.20
- Probability of becoming a Lost customer = 0.05

All other transition probabilities remain unchanged.

Questions:

i) Construct the new transition probability matrix after the introduction of the discount scheme.
ii) Using the new transition matrix, determine the expected time before a Regular buyer becomes a Lost customer, assuming the customer starts in State 1.

5a	A firm allows backorders. D=20,000, S=₹500, H=₹10. Optimal Q* with shortage is 1600. Find shortage cost P?	08	3	3
5b	A manufacturing company uses steel bolts regularly. Its annual requirement is 24,000 bolts. Cost of placing one order is ₹500 and holding cost is ₹12 per bolt per year. Find: (i) EOQ (ii) number of orders per year (iii) time between orders (iv) minimum annual ordering + holding cost	08	3	3
	OR			
6a	Given: batteries/year, purchase price = Rs.28, holding cost rate = 30% ⇒ per unit/year, ordering cost . Current order quantity batteries (per month). Calculate Costs for current order quantity, EoQ, Number of orders per year using EoQ and costs for EoQ	08	3	3
6b	A company needs 26,000 units/year, S=₹150, H=₹5. Find EOQ and time between orders (weeks)	08	3	3
7a	A delivery truck costs ₹90,000 with a scrap value of ₹10,000. The running costs for different years are: Year | 1 | 2 | 3 | 4 | 5 Running Cost (₹) | 6,000 | 9,000 | 13,000 | 18,000 | 24,000 Find the year in which the truck should be replaced to minimize average cost.	08	4	3
7b	A factory purchases a generator for ₹2,00,000. The scrap value at the end of its life is ₹40,000. The yearly operating costs are: Year | 1 | 2 | 3 | 4 Running Cost (₹) | 15,000 | 22,000 | 30,000 | 40,000 Determine the economic life of the generator.	08	4	3
	OR			

<table>
<tr><td>8a</td><td>Two types of bulbs are available for a factory: Type A bulb costs ₹20 and has an average life of 1,000 hours. Type B bulb costs ₹30 and has an average life of 2,000 hours. Electricity and other costs are same. Which bulb is more economical?</td><td>08</td><td>4</td><td>3</td></tr>
<tr><td>8b</td><td>A printing press machine is purchased at a cost of ₹1,60,000 and its scrap value is ₹30,000. The annual running expenses are as follows:

<table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Running Cost (₹)</td><td>14,000</td><td>19,000</td><td>25,000</td><td>32,000</td><td>40,000</td><td>49,000</td></tr></table>
Determine the economic replacement age of the machine.</td><td>08</td><td>4</td><td>3</td></tr>
<tr><td>9a</td><td>The following data pertains to a manufacturing activity.

<table><tr><th>Job, j</th><th>Processing time, t_j</th><th>Weight, w_j</th></tr><tr><td>1</td><td>7</td><td>1</td></tr><tr><td>2</td><td>18</td><td>2</td></tr><tr><td>3</td><td>6</td><td>1</td></tr><tr><td>4</td><td>8</td><td>2</td></tr><tr><td>5</td><td>12</td><td>3</td></tr></table>
Determine the sequence which will minimize the weighted mean flowtime of the above problem. Also find the weighted mean flowtime</td><td>08</td><td>4</td><td>3</td></tr>
<tr><td>9b</td><td>Consider the following two machines and six jobs flow shop-scheduling problem. Using Johnson’s algorithm, obtain the optimal sequence, which will minimize the makespan.

<table><tr><td>Job</td><td>Machine 1</td><td>Machine 2</td></tr><tr><td>1</td><td>5</td><td>4</td></tr><tr><td>2</td><td>2</td><td>3</td></tr><tr><td>3</td><td>13</td><td>14</td></tr><tr><td>4</td><td>10</td><td>1</td></tr><tr><td>5</td><td>8</td><td>9</td></tr><tr><td>6</td><td>12</td><td>11</td></tr></table></td><td>08</td><td>4</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>10a</td><td>Critically examine the assumptions made in sequencing problems with respect to real industrial environments. Suggest practical modifications that can make sequencing models more realistic.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>10b</td><td>Analyze the applicability and limitations of Johnson’s rule in real-world scheduling problems. What happens when the assumptions of Johnson’s rule are violated?</td><td>08</td><td>2</td><td>3</td></tr>
</table>

IM355TBD

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

ADVANCED DECISION MODELING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

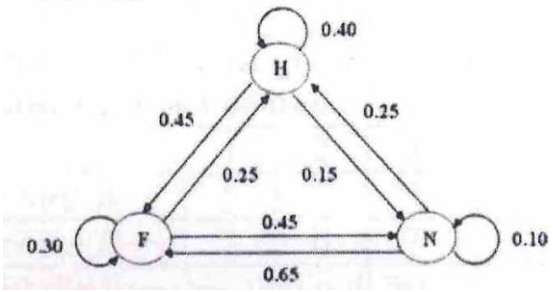
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	Why is queuing theory important in service industries?	02	2	2
	1.2	How does increasing the arrival rate affect the average waiting time in the system?	02	2	2
	1.3	How to the transition rates in a continuous-time Markov chain differ from transition probabilities in a discrete-time Markov chain?	02	2	3
	1.4	What are steady-state probabilities in a Markov chain?	02	2	3
	1.5	What should be considered when applying the EOQ model with quantity discounts?	02	2	2
	1.6	How does the review period affect the deterministic periodic review model?	02	2	3
	1.7	List the key phases of the Bathtub Curve in Replacement and Maintenance Models.	02	2	3
	1.8	What are the two main types of replacement policies?	02	2	2
	1.9	What is the primary objective of sequencing problems? Mention any two.	02	2	2
	1.10	How does the concept of 'setup time' impact sequencing problems?	02	2	2

PART-B

2	a	Explain with a neat diagram, the structure of a queuing system.	06	2	2
	b	In a departmental store one cashier is there to serve the customers. And the customers pick up their needs by themselves. The arrival rate is 9 customers for every 5 minutes and the acshier can serve 10 customers in 5 minutes. Assuming Poisson arrival rate and exponential distribution for service rate, find: i) Average number of customers in the system ii) Average number of customers in the queue or average queue length iii) Average time a customer spends in the system iv) Average time a customer waits before being served.	10	4	4
3	a	What is a discrete-time Markov chain ($DTMC$), and how is it characterized?	08	2	2
	b	State the steps of constructing Matrix of Transition Probabilities of Markov chain.	08	3	3
		OR			

4	a	What is a stochastic process, and how does it differ from a deterministic process?	06	2	2
	b	A professor of statistics not wanting to be predictable, decided on an innovative way of assigning homework based on probabilities. On the first day of the week, he draws a transition diagram as shown in the following Fig. 4.b. The nodes of the diagram represent full credit (F), half credit (H) and no credit (N) assignments. The transition probabilities for the day are as shown in figure. The initial probability distribution of the three state is $(0.3, 0.4, 0.3)$.  Fig. 4.b i) Construct the transition probability matrix ii) $P(X_2 = H)$	10	3	3
5	a	A particular item has a demand of 9000 units per year. The cost of one procurement is $Rs.100$ and the holding cost unit is $Rs.2.40$ per year. The replacement is instantaneous and no shortages are allowed. Determine: i) The economic lot size ii) The number of orders per year iii) The time between orders iv) The total cost per year if the cost of one unit is $Re.1$.	10	3	2
	b	List the limitation of using Economic Order Quantity.	06	2	3
		OR			
6	a	An item is produced at the rate of 50 items per day. The demand occurs at the rate of 25 items per day. If the setup cost is $Rs.100$ per setup and holding cost is $Rs.0.01$ per unit of item per day, find the economic lot size for one run, assuming the shortages are not allowed. Also find the time of cycle and minimum total cost for one run.	08	3	3
	b	A newspaper boy buys paper for 0.05 paise each and sells them for 0.06 paise each. He cannot return unsold newspapers. Daily demand 'r' for newspapers follows the distribution: $Demand\ 'r'$: 10 | 11 | 12 | 13 | 14 | 15 | 16 $Probability\ p(r)$: 0.05 | 0.15 | 0.40 | 0.20 | 0.10 | 0.05 | 0.05 If each day's demand is independent of the previous day's demand, how many papers should be ordered each day?	08	3	3
7	a	Explain the following sudden failure of replacement models with graphs i) Progressive Failure ii) Retrogressive Failure iii) Random Failure	06	2	2

	b	The data collected in running a machine, the cost of which is $Rs.60{,}000$ are given below: $Year$: 1 | 2 | 3 | 4 | 5 $Resale\ value\ (Rs)$: 42,300 | 30,000 | 20,400 | 14,400 | 9,650 $Cost\ of\ spares\ (Rs)$: 4,000 | 4,270 | 4,880 | 5,700 | 6,800 $Cost\ of\ labour\ (Rs)$: 14,000 | 16,000 | 18,000 | 21,000 | 25,000 Determine the optimum period for replacement of Machine.	10	3	3
		OR			
8	a	Discuss the importance of replacement models.	02	2	2
	b	The percentage failure of bulbs at the end of weeks is given below. The cost of replacing individual bulbs $Re.1$. The decision is taken to replace all bulbs simultaneously at fixed intervals and also to replace individual bulbs as they fail in service. If the cost of group replacement is 0.35, find which policy is to be adopted, if initially there are 100 bulbs. $Week$: 1 | 2 | 3 | 4 | 5 $Percentage\ failing\ by\ the\ end\ of\ week$: 10 | 25 | 50 | 80 | 100	14	3	3
9	a	There are five jobs, each of which must go through the two machines A and B in the order AB. Processing times are given below: Job | 1 | 2 | 3 | 4 | 5 $Machine\ A$ | 5 | 1 | 9 | 3 | 10 $Machine\ B$ | 2 | 6 | 7 | 8 | 4 Determine a sequence for the five jobs that will minimize the total elapsed time.	08	3	4
	b	We have five jobs, each of which must go through the machine A, B and C in the order ABC. Determine the sequence that will minimize the total elapsed time. $Job\ No$ | 1 | 2 | 3 | 4 | 5 $Machine\ A$ | 5 | 7 | 6 | 9 | 5 $Machine\ B$ | 2 | 1 | 4 | 5 | 3 $Machine\ C$ | 3 | 7 | 5 | 6 | 7	08	3	4
		OR			
10	a	In the context of manufacturing and production systems, explain role of priority rules related to sequencing problems.	06	2	4
	b	Use the graphical method to minimize time needed to process the following jobs on the machines shown, i.e. for each machine find the job that should be done first. Also, calculate the total elapsed time to complete both jobs. $Machine$ $Job\ 1\ sequence$: A B C D E $Time\ (hrs)$: 3 4 2 6 2 $Machine$ $Job\ 2\ sequence$: B C A D E $Time\ (hrs)$: 5 4 3 2 6	10	3	4

18IM52

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations March / April-2023

Industrial Engineering and Management

DECISION SCIENCES II (STOCHASTIC MODELS)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

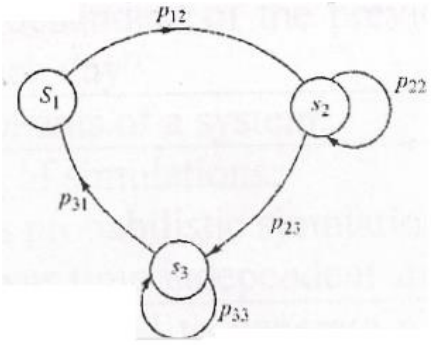
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Table is permitted.

PART-A

1	1.1	List the behavior of arrivals in a queuing system.	02
	1.2	What is static and dynamic queue discipline?	02
	1.3	State the purpose of inventory management.	02
	1.4	List the four components of total inventory cost.	02
	1.5	Probabilities of occurrence of any state are representing one of the _____ numbers of states of nature in the _____.	02
	1.6	What will be the number of equilibrium equations would be, if a matrix of transition probability is of the order $n \times n$?	02
	1.7	The Monte-Carlo method of simulation is developed through the use of _____ and _____.	02
	1.8	Validation is the process of _____ a model to the real system that it represents to make sure that it is _____.	02
	1.9	List the applications, other than systems simulation, for pseudo random generations.	02
	1.10	Is Erlang distribution same as gamma distribution?	02

PART-B

2	a	Explain transient state and steady state condition of a queuing system.	06
	b	A warehouse has only one loading dock manned by a three person crew. Trucks arrive at the loading dock at an average rate of 4 trucks per hour and the arrival rate is Poisson distributed, The loading of a truck takes 10 minutes on an average and can be assumed to be exponentially distributed. The operating cost of a truck is Rs.20 per hour and the members of the loading crew are paid Rs.6 each per hour. Would you advise the truck owner to add another crew of three persons?	10

3	a	State the characteristics of Markov Chain	06
	b	Explain with the diagram shown in Fig 3b, transition probability and transition matrix.  Fig 3b	10
		OR	
4	a	Determine whether or not the following matrices could be a transition matrix for a Markov chain. For those that are not, explain why not, and for those that are, draw a picture of the chain. i) $\begin{vmatrix} 0.5 & 0 \\ 0 & 0.5 \end{vmatrix}$ ii) $\begin{vmatrix} 2/3 & 1 \\ 1/3 & 0 \end{vmatrix}$ iii) $\begin{vmatrix} 0.25 & 0.5 \\ 0.75 & 0.5 \end{vmatrix}$	06
	b	A company is considering using Markov theory to analyze brand switching between three different brands of floppy disks. Survey data has been gathered and has been used to estimate the following transition matrix for the probability of moving between brands each month: $$\begin{array}{cc\|ccc} & & \multicolumn{3}{c}{To\ Brand} \\ & & 1 & 2 & 3 \\ \hline & 1 & 0.80 & 0.10 & 0.10 \\ From\ brand & 2 & 0.03 & 0.95 & 0.02 \\ & 3 & 0.20 & 0.05 & 0.75 \end{array}$$ The current (month1) market shares are 45%, 25% and 30% for brands 1, 2 and 3 respectively. i) What will be the expected market shares after two months have elapsed (i.e. in month 3)? ii) What is the long run prediction for the expected market share for each of the three brands?	10
5	a	Give the type of inventory held for the following type of organization: i) Manufacture ii) Hospital iii) Bank iv) Airline company	06
	b	The production department of a company requires 3,600 kg of raw material for manufacturing a particular item per year. It has been estimated that the cost of placing an order is $Rs.36$ and the cost of carrying inventory is 25 percent of the investment in the inventories. The price is Rs.10 per kg.	

Calculate:

i) The optimal lot size
ii) The optimal order cycle time
iii) Per year minimum total inventory variable cost
iv) Per year minimum total inventory cost — 10

OR

6 a The pattern of demand for a seasonal product is as follows:

Demand (in units)	1	2	3	4	5	6	7	8
Probability	0.05	0.10	0.15	0.20	0.20	0.15	0.10	0.05

The cost of product is Rs.80 per unit and the selling price is Rs.120 How many units should be purchased for the season so as to maximize the expected profit? — 06

b A newspaper boy buys papers for 0.05 paise each and sells them for 0.06 paise each. He cannot return unsold newspapers. Daily demand 'r' for newspapers follows the distribution:

Demand 'r'	10	11	12	13	14	15	16
Probability $p(r)$	0.05	0.15	0.40	0.20	0.10	0.05	0.05

If each day's demand is independent of the previous day's demand, how many papers should be ordered each day? — 10

7 a Explain the different components of a system. — 10

b Discuss the following types of simulations:

i) Deterministic versus probabilistic simulation
ii) Time dependent versus time independent simulation — 06

8 a Use the linear congruential method to generate a sequence of three two-digit random integers and corresponding random numbers. Let $X_o = 27, a = 8, c = 47$ and $m = 100$. — 06

b The sequence of umbers $0.54, 0.73, 0.98, 0.11$ and 0.68 has been generated. Use the Kolmogorov- Smirnov test with $\alpha = 0.05$ to learn whether the hypothesis that the numbers are uniformly distributed on the interval $[0, 1]$ can be rejected. — 10

18IM52

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb – 2021

Industrial Engineering and Management

DECISION SCIENCE II (STOCHASTIC MODELS)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Tables is permitted. (Ks test and ChiSquare tables)

PART-A

1	1.1	In queue designation A/B/S: (d/f), what does S represent?	01
	1.2	The distribution of arrivals in a queuing system can be considered as a _____ process.	01
	1.3	If the operating characteristics of a queue are dependent on time, then it is said to be in a _____ state.	01
	1.4	The characteristics of a queuing model are independent of _____ the queue discipline.	01
	1.5	Write the probability density function of a Markov process.	01
	1.6	In Markov analysis, the likelihood that any system will change from one period to the next is revealed by the matrix of _____ probabilities.	01
	1.7	A problem is classified as Markov chain provided; there are _____ number of possible states.	01
	1.8	The probability that a system is in a particular state after a large number of periods is _____ of the beginning state of the system.	01
	1.9	At Economic Order Quantity, annual ordering cost is _____ to annual carrying cost.	01
	1.10	Carrying cost _____ with quantity ordered per order.	01
	1.11	The ordering cost per order and average unit carrying cost are constant, and demand suddenly falls by 75 % then EOQ will _____ by 50 percent.	01
	1.12	The water consumption from a water tank follows _____ system.	01
	1.13	If EOQ = 40 units, order costs are Rs. 2 per order, and carrying costs are Rs. 20 per unit, what is the usage in units?	01
	1.14	The cost of insurance and taxes are included in _____ carrying cost.	01
	1.15	The time period between placing an order and its receipt in stock is known as _____ time.	01
	1.16	The purpose of using simulation technique is to _____ a real world situation.	01
	1.17	Biased random sampling is made from among alternatives which have _____ probability.	01

	1.18	A simulation uses the logical relationships and mathematical expressions of the _____ system.	01
	1.19	_____ model can only show the values that system attributes take when the system is in balance.	01
	1.20	From a practical perspective, if we have a waiting line problem for which the Poisson and negative exponential distributions do not apply, and we desire a reasonably accurate solution, we should use _____.	01

PART-B

2	a	Discuss the characteristics of a queuing system.	08
	b	A super market has a single cashier. During the peak hours, customers arrive at a rate of 20 customers per hour. The average no of customers that can be processed by the cashier is 24 per hour. Find (i) The average no of customers in the queue system (ii) The average time a customer spends in the system. (iii) The average time a customer spends in queue. (iv) The any time a customer spends in the queue waiting for service	08
3	a	State and discuss the characteristics of a Markov Chain.	06
	b	State the steps of constructing Matrix of Transition Probabilities of Markov Chain.	10
		OR	
4	a	State the application of Markov analysis.	04
	b	A food products' company is contemplating the introduction of a revolutionary new product with packaging or replacing the existing product at much higher price (S1). It may even make a moderate change in the composition of the existing product, with a new packaging at a small increase in price (S2), or may make a small change in the composition of the existing product, backing it with the word 'New' and a negligible increase in price (S3). The three possible states of nature or events are (i) high increase in sales (N1), (ii) no change in sales (N2) and (iii) decrease in sales (N3). The marketing department of the company worked out the payoffs in terms of yearly net profits for each of the strategies of three events (expected sales). This is represented in the following table:	12

		States of Nature		
		N1	N2	N3
Strategies	S1	7,00,000	3,00,000	1,50,000
	S2	5,00,000	4,50,000	0
	S3	3,00,000	3,00,000	3,00,000

Which strategy should the concerned executive choose on the basis of
i. Maximin criterion
ii. Maximax criterion
iii. Minimax regret criterion
iv. Laplace criterion

5 a List the assumptions of economic order quantity model. 06

b ABC Ltd. uses EOQ logic to determine the order quantity for its various components and is planning its orders. The Annual consumption is 80,000 units, Cost to place one order is Rs. 1,200, Cost per unit is Rs. 50 and carrying cost is 6% of Unit cost. Find

i. Economic Order Quantity
ii. No. of orders per year
iii. Ordering Cost
iv. Carrying Cost
v. Total Cost of Inventory. 10

OR

6 a Explain the following cost associated with inventories:

i. Purchase cost
ii. Carrying cost
iii. Ordering cost 06

b A contractor of second hand motor trucks use to maintain a stock of trucks every month. The demand of the trucks occur at a constant rate but not in constant size. The probability distribution of the demand is as shown below:

Demand	0	1	2	3	4	5	6 or more
Probability	0.40	0.24	0.20	0.10	0.05	0.01	0.00

The holding cost of an old truck in stock for one month is /Rs.100/- and penalty for a truck if not delivered to the demand, is Rs. 1000/-. Determine the optimal size of the stock for the contractor. 10

7 a Explain at least four different types of model. 08

b Using Multiplicative Congruential Method, generate sequence of 5 integer numbers where a = 24, m = 100 and seed value = 64. 08

8 a Using KS test $\alpha = 0.05$, to test whether the data shown below are uniformly distributed.
0.44, 0.81, 0.14, 0.05, 0.93 08

b Using Chi - Square test with level of significance $\alpha = 0.05$, test whether the data shown below are uniformly distributed. With tests using 10 intervals of equal length and total number of sample data are 100.

Demand	1	2	3	4	5	6	7	8	9	10
Observations	8	8	10	9	12	8	10	14	10	11

08